Fine-Tuning Creates Confident Errors: Clarification Failure in Bengali–English Agricultural Question Answering

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Ethics Statement

Human participants

The only people who took part in this study were the raters of model outputs. Five annotators rated 800 model outputs, giving 1200 ratings in total. We believe the five annotators are the five authors of this thesis; each author's confirmation and background are **AUTHOR INPUT:** confirm annotators and their backgrounds. The study design also includes two external raters who are not members of the team and who rate a shared subset of 100 outputs; their ratings belong to a supplementary round that had not run when this thesis was compiled (**STAGE 8B, NOT YET RUN:** external raters). No farmer, no other member of the public and no minor took part. Raters saw questions and model answers about crop irrigation only. No personal data were collected: ratings are stored under the codes A1 to A5.

Consent

Written consent was not collected before the authors did their ratings. A consent form now exists. For the authors it records, truthfully and with the real signing date, that the rating was completed before the form was signed; external raters sign it before they rate. The signed forms are **PENDING CONSENT:** signed consent forms. Whether the department required formal ethics approval for rating by the authors themselves is **AUTHOR INPUT:** supervisor's answer on ethics approval.

Data and licences

The domain facts come from the crop calendars published by the Bangladesh Agricultural Research Council (BARC) through its online crop-calendar application [1]; **AUTHOR INPUT:** BARC terms of use: research use, data release. The irrigation calculation follows the method of FAO Irrigation and Drainage Paper 56 [2]. The four models were used under their published licences (Section 3.10). All questions and answers were written or generated for this study; none came from real users.

Deployment and risk

The system described here is a research testbed. It was never deployed and no farmer received its advice. Its fine-tuned models invent missing inputs and present the result with confidence (Chapter 5); advice of that kind could lead a farmer to over- or under-irrigate. We therefore do not recommend deploying any of the models studied here, and we report every limitation we know of in Section 5.10.

Use of AI tools

AI assistants were used in this project for writing and checking code, for statistical scripting and for drafting and revising text; every number in the thesis is generated from the project's files by scripts, and every reference was checked against its source. The declaration required by the department is **AUTHOR INPUT:** department's AI-declaration wording; the full declaration is in Chapter H.

Abstract

Fine-tuning is widely used to adapt large language models to specialised domains, and is typically evaluated on task accuracy alone. We show that this evaluation misses a failure mode that fine-tuning creates.

We built a bilingual Bengali–English question-answering system for irrigation advisory, targeting runtime-conditioned queries whose answers depend on live evapotranspiration, rainfall, crop stage and field size, and therefore cannot be retrieved from any document. Four models (Llama-3.1-8B, Mistral-NeMo-12B, Phi-3.5-mini, Phi-4-mini) were fine-tuned with three seeds each on a leakage-audited corpus derived from BARC crop calendars, and evaluated on a hardened 422-item test set (362 items scored) containing adversarial misspellings, unit ambiguity and incomplete queries. Five annotators rated 800 model outputs, and all hypotheses were tested under an analysis plan fixed before any significance test was computed.

Fine-tuning taught the models the system's conventions, which no base model possessed. All four scored 0.000 F1 on the crop-coefficient table and on the tiered effective-rainfall rule before training, and 0.542–0.761 and 0.930–0.999 respectively after. Accuracy within ±0.1 mm on single-crop queries rose from 0.0–3.4% to 38.8–69.3%, a gain that held in every model and seed.

The same training produced no clarification behaviour in any of twelve training runs. When a required parameter was absent, every fine-tuned model supplied a value the user never gave and computed with it, while refusing out-of-scope questions correctly; abstention appeared in the training corpus and clarification did not. Base models also invented values, but often visibly: 13.2–66.2% of their responses to incomplete queries ran to the token limit, and 15–42% of their invented answers said a value had been assumed. After fine-tuning, 98.5–100% were complete, unhedged, confident errors in every model, and annotators judged none of the evaluated fine-tuned models' responses to incomplete queries fit to give a farmer.

Adding 300 clarification examples raised clarification that names the missing parameter to 29.4–38.2% and cut invented values from 100% to 50.0–66.2%, with no detectable loss of numeric accuracy. Bengali trailed English on factual text in all four models (6.4–12.8 chrF++ points), significantly in three; differences in numeric accuracy and in human ratings between the languages were not significant.

Keywords: large language models; fine-tuning; clarification; abstention; hallucination; Bengali; agricultural question answering; irrigation advisory

Acknowledgement

AUTHOR INPUT: Acknowledgement text, written by the authors (supervisor, co-supervisor, anyone who helped, and the external raters if they agree to be named).

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List of Abbreviations and Symbols

The following list describes the abbreviations and symbols used in the body of the document.

Abbreviations

AUC Area under the receiver operating characteristic curve

BARC Bangladesh Agricultural Research Council

bn Bengali (language code)

chrF++ Character *n*-gram F-score with word unigrams and bigrams

CI Confidence interval (95 % bootstrap unless stated)

CL Clarify-trained model (fine-tuned on the corpus plus clarification examples)

CUDA Compute Unified Device Architecture (NVIDIA GPU platform)

en English (language code)

FAO Food and Agriculture Organization of the United Nations

FT Fine-tuned model

GPU Graphics processing unit

IEEE Institute of Electrical and Electronics Engineers

LLM Large language model

LoRA Low-rank adaptation

MAE Mean absolute error

NF4 4-bit NormalFloat quantisation

NLP Natural language processing

OR Odds ratio

pp Percentage points

QA Question answering

QLoRA Low-rank adaptation on a quantised base model

RAG Retrieval-augmented generation

RMSE Root mean squared error

ROUGE-L Recall-Oriented Understudy for Gisting Evaluation, longest common subsequence

SFT Supervised fine-tuning

SHA-256 Secure Hash Algorithm, 256-bit digest

TGP Total graphics power

VRAM Video memory of the GPU

ZS Zero-shot (base model, no fine-tuning)

Symbols

α Krippendorff's alpha (inter-rater reliability)

Δ Difference between two arms (treatment minus baseline)

κ Fleiss' kappa (inter-rater agreement)

ET_0 Reference evapotranspiration (mm per day)

ET_c Crop evapotranspiration, $K_c \times \mathrm{ET}_0$ (mm per day)

ρ Spearman rank correlation

K_c Crop coefficient

P_{eff} Effective rainfall (mm)

p_{Holm} p-value after Holm correction within a test family

r_{rb} Rank-biserial correlation (effect size of the Wilcoxon test)

W Kendall's coefficient of concordance (effect size of the Friedman test)

Chapter 1

Introduction

1.1 Background

Agriculture in Bangladesh depends on irrigation in the dry season. Millions of smallholder farmers pump groundwater to grow dry-season rice and other crops, and the combined effect of their pumping is visible at the scale of the whole Bengal Basin [3]. Each of these farmers faces the same small decision many times a season: how much water should this field receive today?

The standard way to answer that question is the crop-coefficient method of the Food and Agriculture Organization of the United Nations (FAO), published as FAO Irrigation and Drainage Paper 56 [2]. The method multiplies the reference evapotranspiration of the day, ET_0, by a crop coefficient, K_c, that depends on the crop and its growth stage. The product is the crop's water use, ET_c. Subtracting the part of recent rainfall that the soil can use, the effective rainfall P_{eff}, gives the net depth of water to apply. Multiplying the depth by the field area gives a volume. Four of these inputs change from day to day or from farm to farm: ET_0, rainfall, the growth stage and the field size. Only the crop coefficient and the crop's calendar are fixed facts that can be written down once.

In Bangladesh, crop calendars are published by the Bangladesh Agricultural Research Council (BARC) through an online application [1]. A calendar says when a crop is sown, when each growth stage begins and when irrigation is critical. It cannot say how much rain fell on a particular field yesterday, or how large that field is. We call a question whose answer depends on such live values a *runtime-conditioned query*. The answer to "how much should I irrigate my Boro rice field today?" is not stored in any document. It has to be computed from values that only the farmer, or a weather service, can supply at the moment of asking.

Large language models (LLMs) are now being built into agricultural advisory services. Farmer.Chat answers farmers' questions through messaging apps in several countries by retrieving from a curated knowledge base [4]. KrishokBondhu offers voice-based advice in Bengali to Bangladeshi farmers [5]. Other systems compare retrieval with fine-tuning for agricultural questions [6] or connect farmers to practice guides through retrieval-augmented generation (RAG) [7]. A review in *Nature Food* argued that such systems could widen access to extension advice but need experts in the loop to keep the advice safe [8]. Retrieval can supply the calendar and the coefficient table. It cannot supply today's rainfall

or the size of a farmer's field. For a runtime-conditioned query, those values must come from the question itself, and a system that answers such questions must notice when one of them is missing.

1.2 Rationale of the Study or Motivation

Fine-tuning is the usual way to adapt a general LLM to a narrow domain, and parameter-efficient methods such as low-rank adaptation (LoRA) [9] and its quantised variant QLoRA [10] make it cheap enough for a student project on a single graphics processing unit (GPU). A fine-tuned model is then judged by how often it gives the right answer. Accuracy is a natural measure, but it is incomplete. Kalai and colleagues argue that when training and evaluation reward only correct answers, a model that guesses scores better than one that admits it does not know, so accuracy-only evaluation pushes models towards confident guessing [11]. Gekhman and colleagues showed that fine-tuning on examples that introduce new knowledge makes a model more likely to hallucinate [12].

An irrigation assistant faces a sharper version of this problem. A farmer who forgets to mention the field size has asked a question with no single correct answer. The right response is to ask for the missing value. A model trained only on complete questions has never seen that response. If it fills the gap with a plausible number and returns a clean, well-formatted result, the farmer has no signal that anything went wrong. A base model that rambles, trails off or says it is guessing is also wrong, but its failure is visible. We are interested in this difference in the form of the error, and we define it as follows.

> We define a confident error as a response that supplies a value the user never provided and delivers it as a complete, well-formed final answer, without truncation and without signalling that anything was assumed. Under this definition the question is not whether a model errs when information is missing, since base models do as well, but whether fine-tuning changes the form of that error from visibly unreliable to authoritative.

This definition matters for safety. An irrigation depth that is too high wastes pumped groundwater and money; one that is too low can stress the crop at a sensitive stage. An error that looks authoritative is more likely to be acted on than one that looks broken.

1.3 Problem Statement

Fine-tuned LLMs for agricultural advice are evaluated mainly by the accuracy of their answers to complete questions. That evaluation does not show what a fine-tuned model does when a question lacks an input the answer depends on, and it does not distinguish an error the user can see from one the user cannot. For Bengali–English irrigation advice in Bangladesh, where questions are runtime-conditioned, we do not know whether standard fine-tuning teaches a model to ask for a missing value, to refuse, or to invent one, nor whether it changes how that invented value is presented. We also do not know whether a small amount of targeted training data can change this behaviour, or how the answer differs between Bengali and English.

1.4 Objectives

The aim of this thesis is to measure, in a controlled setting, what LoRA fine-tuning teaches LLMs about incomplete irrigation questions, and whether it changes the form of their errors. We built a bilingual question-answering (QA) testbed from BARC crop calendars, fine-tuned four open-weight models on it, and evaluated them on a hardened test set with automatic metrics, a human rating study and significance tests under an analysis plan fixed before any significance test was computed. The research questions and the hypotheses tested for each are listed below.

RQ1. Does fine-tuning on complete-input examples teach the models the system's conventions (its crop-coefficient table and effective-rainfall rule) and its arithmetic? *H1: fine-tuned models are more accurate than their zero-shot base models on numeric questions.*

RQ2. When a required input is missing, do fine-tuned models ask for it, refuse, or invent it, and does fine-tuning change the form of the error? *T (the title claim): fine-tuning raises the rate of confident errors, as defined in Section 1.2, on incomplete queries.*

RQ3. Can a few hundred templated clarification examples teach the models to ask, and at what cost to numeric accuracy? *C1: clarify training raises clarification that names the missing parameter. C2: it lowers the rate of invented values. C3: it does not reduce numeric accuracy.*

RQ4. Does performance differ between Bengali and English? *H2: Bengali trails English after fine-tuning.*

RQ5. Do human raters agree with one another, and do their judgements corroborate the automatic measures? *H3: agreement on the primary human measures reaches a pre-specified threshold.*

The clarification category in the test set is a designed probe. We added incomplete questions to the test set knowing that the training corpus contained no clarification examples, because the purpose was to observe what fine-tuning does in their absence. Its absence from training is part of the design, not a finding. The finding is what the models did instead, and how that compares with abstention, which was present in training.

1.5 Methodology in Brief

Fig. 1.1 summarises the stage-gated pipeline. Each stage ended with a written gate report that had to pass before the next stage began. We digitised 28 BARC crop calendars into 902 seed question–answer rows covering 451 facts, half in Bengali and half in English (stage S0). We split these rows into training, validation and test sets by source group before any augmentation (S1), so that no fact family crossed the split. Only the training split was augmented, to 3552 rows (S2), and four leakage detectors compared the augmented training rows with the held-out rows (S3). The test split was then hardened with 240 adversarial rows, including misspelled crops, ambiguous units and questions with a missing parameter, and locked with a SHA-256 hash (S4).

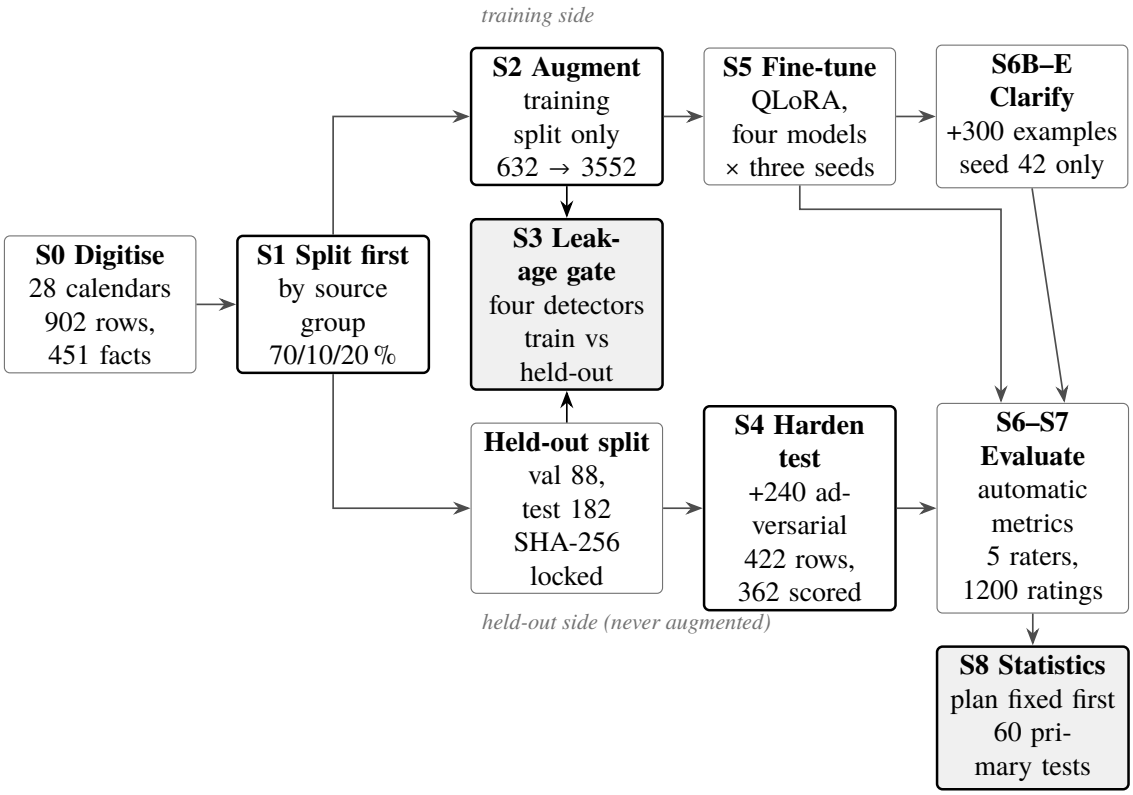


Figure 1.1: The stage-gated pipeline (S0–S8). Boxes with a heavy border are the stages that protect the test set: splitting by source before augmentation (S1), augmenting only the training side (S2), the four-detector leakage gate between the two sides (S3) and the hashed, hardened test set (S4). Counts are rows. The corpus was split before any augmentation, so no augmented row can share a source with a test row.

We fine-tuned Llama-3.1-8B, Mistral-NeMo-12B, Phi-3.5-mini and Phi-4-mini with QLoRA, each with seeds 42, 1337 and 2024, giving twelve training runs (four models × three seeds) (S5). We evaluated all runs and the four base models on the locked test set with metrics gated by question type (S6). We then retrained each model at seed 42 with 300 added clarification examples (S6B–E). 5 annotators rated 800 outputs from four arms (S7), and all hypotheses were tested with paired tests and Holm correction under a plan fixed before significance testing (S8).

1.6 Scope and Challenges

The study has deliberate limits. Of the 422 rows in the hardened test set, 362 were scored. The other 60 rows (40 adversarial out-of-domain questions and 20 questions with conflicting values) have no machine-checkable gold answer and were excluded by design. Abstention is therefore measured on the 34 out-of-domain rows of the seed test split only, and behaviour on conflicting values is not evaluated. All questions are synthetic or templated from calendar facts; none was asked by a real farmer. We studied four open-weight models, the largest being Mistral-NeMo-12B, and no commercial model.

Retrieval is out of scope. The original plan agreed with the supervisor was to fine-tune a model, compare it with retrieval, and fall back to retrieval when the model could not

answer. The version reported here dropped the retrieval arm, because retrieval cannot supply the live inputs that runtime-conditioned queries depend on, and because the behaviour we study, what a model does when an input is missing, has to be solved before any retrieval or calculator can help. The supervisor's agreement with this change is **AUTHOR INPUT:** supervisor's written agreement on dropping the RAG arm.

The project also met practical challenges, each of which changed the design. The second version of the system (v2) fine-tuned one model on an augmented corpus built from the same facts as its test questions, so reworded versions of the test facts were present in training. It scored a macro-F1 of 0.999 across six input slots on 180 held-out numeric rows, a result that measured memorisation rather than learning. The version reported here (v3) was redesigned around that failure: split first, augment only the training side, audit for leakage, and evaluate several models and seeds. During evaluation we found and fixed several faults before they could distort results, among them a Bengali text normaliser that destroyed vowel signs and decimal points, a join that silently dropped seed rows, a ROUGE-L implementation that deleted all Bengali text, and a response classifier that mistook a courtesy caveat for a clarification request. Three metric definitions were changed after results were seen; each change was checked against a gold-oracle arm and is reported in Section 4.4.3.

1.7 Contributions

This thesis extends prior work on abstention, clarification and hallucination in fine-tuned LLMs to a low-resource, bilingual and domain-specific setting. Its contributions are three.

- An abstain-versus-clarify contrast inside the same training runs. In all twelve training runs across four architectures, abstention, which made up 3.6% of the training corpus, was learned completely, while clarification, which made up 0.0%, was not learned at all. This extends benchmark evidence that models fail to ask for missing information [13], [14] to the effect of fine-tuning itself.
- A measurement of the form of the error, not only its rate. Base models also invented missing values, but often visibly, by running to the token limit or saying that a value was assumed. After fine-tuning, almost every invented value came as a complete, unhedged answer. We test this change directly (test A7) under a definition fixed in advance of testing.
- A leakage-audited Bengali–English testbed with human evaluation and a fixed analysis plan. The test set is split by source before augmentation, hashed and hardened with adversarial rows; 800 outputs were rated by 5 annotators; and 60 primary hypotheses were tested with Holm correction under a plan written before any significance test was computed. We also show that 300 templated clarification examples partly correct the failure, with no detectable loss of numeric accuracy.

1.8 Thesis Organisation

Chapter 2 introduces the irrigation method, the models and the evaluation methods the thesis relies on, and reviews work on agricultural LLMs, hallucination, abstention, clarification, Bengali language processing, leakage and human evaluation. Chapter 3 sets out the requirements of the study and its societal, environmental, ethical and economic aspects. Chapter 4 describes the data, the pipeline, training, evaluation, the clarify-training intervention, the human study and the analysis plan. Chapter 5 reports the results, the statistical tests and the threats to validity. Chapter 6 answers the research questions and proposes future work. The appendices hold examples, per-seed and per-category results, the rating instructions, the consent form, the detector rules, the reproducibility record and the declaration of AI assistance.

Chapter 2

Literature Review

This chapter first introduces the background the thesis relies on: the irrigation method, the crop calendars, the language models and how they are adapted, the metrics, and the statistics (Section 2.1). It then reviews research on agricultural language models, hallucination, abstention, clarification, Bengali language processing, test-set leakage and human evaluation (Section 2.2), and closes with what that research establishes and what it leaves open (Section 2.3).

2.1 Preliminaries

2.1.1 Crop water requirement and the FAO-56 method

FAO Irrigation and Drainage Paper 56 gives the standard procedure for computing how much water a crop uses [2]. It separates the effect of weather from the effect of the crop. The weather enters through the reference evapotranspiration, ET_0: the water that a standard, well-watered grass surface would lose to the air on a given day, in millimetres. The crop enters through a crop coefficient, K_c, which scales ET_0 to the crop's own water use, $ET_c = K_c \times ET_0$. In the single-coefficient approach, K_c changes over the season: it is low while the crop is young, highest in the middle of the season and falls again as the crop matures. FAO-56 tabulates typical values for the initial, mid-season and late-season stages of many crops, and recommends adjusting them for local climate.

A field needs irrigation to make up whatever part of ET_c rain has not supplied. Only part of the rain that falls is available to the crop, because some runs off, some drains below the roots and some evaporates from the surface. The available part is the effective rainfall. FAO Irrigation and Drainage Paper 25 discusses its definitions and the many methods of estimating it, from soil-water measurements to empirical rules [15]. Empirical rules typically apply a smaller fraction to heavy rain than to light rain. The testbed in this thesis uses such a rule with three tiers ((4.2)). That rule, its thresholds and its fractions are the project's own convention: they are not taken from FAO-56, FAO-25 or BARC, and a model can only know them if it has seen examples of them.

2.1.2 Crop calendars and irrigation in Bangladesh

Dry-season irrigation in Bangladesh is supplied mostly by groundwater. Rising withdrawals have enabled dry-season rice cultivation across the Bengal Basin, and the collective pumping of millions of smallholder farmers has measurably changed how much of the monsoon the aquifers capture [3]. Irrigation decisions are therefore made field by field, by farmers who each choose when to run a pump and for how long.

The Bangladesh Agricultural Research Council publishes crop calendars through an online application [1]. For each crop, a calendar gives the sowing or planting window, the growth stages and their timing, the periods when irrigation matters most, and notes on fertiliser and pests. These are fixed facts that a question-answering system can store. The inputs that change daily or from farm to farm, such as ET_0, rainfall and the size of a particular field, are not in any calendar.

2.1.3 Large language models and instruction tuning

The models used in this thesis are decoder-only Transformers [16], pre-trained to predict the next token of text and then tuned to follow instructions. Instruction tuning on many tasks written as natural-language instructions improves a model's performance on tasks it was not trained on [17], and further tuning with human feedback makes models follow the intent of a user's request more closely [18]. The four checkpoints studied here are instruction-tuned releases of the Llama 3 [19], Mistral NeMo [20], Phi-3.5 [21] and Phi-4 [22] families. Each has its own tokenizer, which splits text into subword units; the same sentence can take very different numbers of tokens in different languages (Section 2.2.5).

2.1.4 Parameter-efficient fine-tuning

Updating every weight of a model with billions of parameters needs more memory than a single consumer GPU has. Parameter-efficient methods train a small number of new parameters and leave the pre-trained weights frozen. Adapter modules insert small trainable layers into each Transformer block [23]. Low-rank adaptation (LoRA) instead adds to selected weight matrices a trainable update that is the product of two low-rank matrices, so that the update has far fewer parameters than the matrix it modifies [9]. QLoRA goes further: the frozen base model is stored in 4-bit NormalFloat (NF4) precision and gradients flow through it into the LoRA adapters, which makes it possible to fine-tune models of this size on one GPU [10]. This thesis uses QLoRA with adapters on every projection matrix (Section 4.2.3).

2.1.5 Retrieval-augmented generation and runtime values

Retrieval-augmented generation (RAG) combines a language model with a searchable document index: for each question, a retriever finds relevant passages and the model generates an answer conditioned on them [24]. RAG suits questions whose answers are written down somewhere, such as a crop's sowing window. It cannot supply a value that exists in no document. For a runtime-conditioned irrigation query, the day's ET_0, the recent rainfall

and the size of the farmer's field must come from the question, a sensor or a weather service. A retrieval system therefore faces the same problem as a fine-tuned one when the farmer leaves one of those values out: something has to notice that it is missing.

2.1.6 Evaluation metrics

For numeric answers the natural measure is whether the stated value is correct within a tolerance, reported as the share of rows that meet it. For free-text answers there is rarely one correct wording, so overlap metrics are used. chrF scores the overlap of character n-grams between an answer and a reference as an F-score [25], and chrF++ adds word unigrams and bigrams [26]. Character-level metrics suit a language such as Bengali, whose words take many inflected forms. ROUGE-L scores the longest common subsequence of words [27]. BERTScore compares the contextual embeddings of the tokens in an answer and a reference [28]. Word-level Dice, twice the number of shared words divided by the total number of words, is the simplest of these. Automatic overlap metrics agree only weakly with human judgements of generated text [29], which is why this thesis checks them against human ratings (Section 5.7).

2.1.7 Inter-rater agreement

When several people rate the same items, their agreement shows how much a rating reflects the item rather than the rater. Fleiss' κ measures chance-corrected agreement among many raters assigning items to nominal categories [30]. Krippendorff's α handles any number of raters, missing ratings and different levels of measurement, including ordinal scales, and has been proposed as a standard reliability measure [31]. Both equal zero at chance and one at perfect agreement. Conventional bands for interpreting agreement coefficients exist [32], but we report the coefficients with their confidence intervals and a threshold named before testing rather than rely on verbal labels.

2.1.8 Paired significance tests and multiple comparisons

When two systems answer the same test items, their results are paired and a paired test is more powerful than an unpaired one. For binary outcomes (correct or not), McNemar's test uses only the discordant items, those that one system gets right and the other wrong [33]. For continuous or ordinal scores, the Wilcoxon signed-rank test ranks the paired differences [34]. For more than two systems on the same items, the Friedman test compares ranks within each item [35], and Cochran's Q does the same for binary outcomes [36]. The bootstrap estimates the uncertainty of any statistic by resampling the items with replacement [37]; we use it for confidence intervals of differences. Running many tests raises the chance of a false positive; Holm's procedure controls this by comparing the ordered p-values with thresholds that relax step by step [38]. A survey of significance testing in natural language processing (NLP) found that it is often ignored or misused, and proposed a protocol for choosing tests [39].

2.2 Review of Existing Research

2.2.1 Language models for agricultural advisory

Several groups have studied LLMs for agricultural extension. Tzachor and colleagues assessed whether a general-purpose model could simplify scientific knowledge and give personalised, location-specific advice, tested it on technical advice for cassava farmers in Nigeria, and concluded that safe use needs a design process with human experts in the loop [8]. Silva and colleagues evaluated LLMs on agriculture exams from Brazil, India and the United States, and found that GPT-4 answered enough questions to pass agronomist certification exams [40]. Benchmarks such as AgXQA were built specifically for agricultural extension questions [41].

Systems for farmers have mostly used retrieval. Farmer.Chat is a generative-AI chatbot deployed in four countries that draws on a curated knowledge base [4], and AgroLLM compares three LLMs inside a RAG pipeline over agricultural texts [7]. Balaguer and colleagues compared RAG with fine-tuning for location-specific agricultural questions and found that each helped and that the gains added up [6]. For Bangladesh, Krishok-Bondhu is a voice-based advisory service for Bengali-speaking farmers built on RAG over agricultural handbooks and extension manuals [5], and a cross-lingual pipeline translates Bengali questions into English to retrieve from English manuals [42]. KrishokChat is a Bengali agricultural benchmark built from government publications with a separate set of real farmer queries; its authors report that fine-tuning reduced refusal on safety-critical queries [43].

Two features separate this literature from the present study. First, its questions are knowledge questions: what to plant, which pesticide to use, what a symptom means. The answers exist in documents. Second, its evaluations ask whether answers are correct, relevant or well grounded, not what a system does when the question lacks information the answer depends on. The KrishokChat finding that fine-tuning reduced refusal points in the same direction as the present study, but in a different setting.

2.2.2 Hallucination and fine-tuning

Hallucination, text that is fluent but unsupported or wrong, was surveyed first for natural language generation in general [44] and then for LLMs, where plausible but non-factual content is the main concern [45]. Several studies connect it to fine-tuning. Gekhman and colleagues controlled how many fine-tuning examples introduced knowledge the model did not already have; such examples were learned more slowly than the others, and once learned, they increased the model's tendency to hallucinate roughly in proportion to their number [12]. Kang and colleagues found that a fine-tuned model's hallucinated answers to unfamiliar questions mirror the answers attached to the unfamiliar examples in its fine-tuning data, so changing how those examples are supervised changes the behaviour [46]. R-Tuning starts from the observation that ordinary instruction tuning makes a model complete an answer whether or not it knows it, and teaches models to decline questions outside their knowledge instead [47].

Kalai and colleagues give a statistical account of why the behaviour persists. Training

and evaluation reward guessing over admitting uncertainty, because a model graded on accuracy alone scores better by guessing, so benchmarks that ignore uncertainty keep encouraging confident answers [11]. Kadavath and colleagues showed that large models are, when asked in the right format, reasonably well calibrated about whether they know an answer [48]; the problem is less that a model cannot tell when it is uncertain than that its training does not reward saying so.

These studies concern facts a model does not know. The present study concerns a different gap: a value that no model could know, because the user did not provide it. The mechanism these papers describe, that fine-tuning teaches a model to produce the kind of answer its training examples contain, is the one our results point to.

2.2.3 Abstention and selective prediction

A system abstains when it declines to answer. Reading-comprehension benchmarks have required abstention on questions whose answer is not in the passage [49], and selective question answering asks a model to answer as many questions as it can while abstaining where it is likely to be wrong, including under a shift in domain [50]. A survey of abstention in LLMs organises the literature by the query, the model and human values [51]. Brahman and colleagues widened the scope of noncompliance beyond unsafe requests to incomplete, unsupported and indeterminate ones, found that models comply with many requests they should not, and reported that training with low-rank adapters balanced appropriate refusal against other abilities better than full fine-tuning [52].

Two recent benchmarks study questions that are missing information. AbstentionBench evaluates abstention across many datasets, including underspecified questions, and finds that abstention is unsolved, that fine-tuning for extended step-by-step problem solving (so-called reasoning models) degrades it, and that a carefully written system prompt helps in practice without solving the problem [14]. Fan and colleagues gave models maths questions with a missing premise; models tuned for long chains of thought wrote much longer answers and failed to abstain, while models without that tuning noticed the missing premise more often [53]. Both studies show that a kind of fine-tuning can make models worse at recognising an unanswerable question. Neither separates abstention from clarification, and neither concerns a domain where the missing value is a routine input.

2.2.4 Clarification questions and underspecified inputs

Asking for clarification is the alternative to guessing and to refusing. Rao and Daumé ranked candidate clarification questions by how useful their answers would be [54], and Aliannejadi and colleagues showed that a single good clarifying question can greatly improve retrieval in conversational search [55]. In open-domain question answering, over half of the questions in one benchmark turned out to be ambiguous [56].

With generative LLMs the question became whether models ask on their own. Kuhn and colleagues found that current models rarely ask users to clarify ambiguous questions and give incorrect answers instead; prompting a model to detect ambiguity and ask first helped [57]. Zhang and Choi built a framework for deciding when a system should ask, based on the spread of plausible user intents [58]. Several studies train the behaviour in.

Zhang, Knox and Choi traced the problem to preference data that judges responses only by their prior context, and taught models to ask by labelling preferences with simulated future turns [59]. STaR-GATE fine-tuned a model iteratively on the questions that most improved its later answers [60]. ClarifyCoder fine-tuned code models on synthetic examples in which an ambiguous programming task calls for a question before any code [61].

QuestBench is the closest to our clarification probe. It poses reasoning problems with exactly one missing variable and asks whether a model can pick the one question that would resolve it; models struggled to identify the right question even when they could solve the fully specified version [13]. Our incomplete irrigation questions have the same structure: one required input is absent and one question would supply it. What we add is the effect of fine-tuning on this behaviour, a comparison with abstention inside the same training runs, and a measurement of the form the error takes.

2.2.5 Bengali language processing and tokenizer cost

Most of the world's languages are barely represented in language technology [62]. Bengali is widely spoken but has long been treated as a low-resource language in NLP; BanglaBERT and its benchmark were built to address that for language understanding [63]. Evaluations of LLMs on Bengali tasks found that zero-shot models performed poorly on most tasks, and open-weight models especially so [64], and a recent study of open models on translated benchmarks confirmed a substantial English–Bengali gap, part of which comes from answer-format failures that differ between the languages [65].

Tokenisation adds a cost. The same text translated into different languages can take very different numbers of tokens [66], so speakers of some languages pay more per request, fit less text into a context window and often get poorer results [67]. The same study of open models found that Bengali costs several times more tokens per word than English, with the factor set by the tokenizer rather than the script [65]. We measure this cost for our four tokenizers (Fig. 4.3) because it affects both training and the comparison of validation losses.

2.2.6 Leakage and contamination in evaluation

A test score means little if the test items, or close variants, were seen in training. Lewis, Stenetorp and Riedel found that a large share of test questions in popular open-domain QA benchmarks had near-duplicate paraphrases in the training sets, and that models did much worse on questions they could not have memorised [68]. Pre-trained models can exploit test data present in their pre-training corpus [69], contamination makes models look better than they are and should be measured for every benchmark [70], and methods now exist to detect it [71]. Outside NLP, a survey of machine-learning-based science found data leakage to be widespread and a cause of severe failures to reproduce published results [72]. These studies motivated the split-first design and the four leakage detectors of this thesis, and the lesson of the second version of our own system (Section 1.6).

Table 2.1: Related work compared on the dimensions this thesis studies. Entries record only what each work's abstract states; "–" means the abstract does not state it, not that the work lacks it. "Missing-input probe" means the evaluation includes questions that cannot be answered without information the user did not give.

Work	Domain	Language	Fine-tuning studied	Missing-input probe	Human evaluation
QuestBench [13]	logic, planning, maths	en	–	yes (one missing variable)	–
AbstentionBench [14]	general QA	en	yes (reasoning-tuned vs base)	yes (underspecified)	–
Missing premise [53]	maths	en	yes (reasoning-tuned vs base)	yes (missing premise)	–
Gekhman et al. [12]	closed-book QA	en	yes	no (unknown facts)	–
Kang et al. [46]	QA, biographies	en	yes	no (unfamiliar queries)	–
R-Tuning [47]	QA	en	yes	no (unknown facts)	–
CLAM [57]	ambiguous QA	en	no (prompting)	yes (ambiguity)	–
STaR-GATE [60]	personalised tasks	en	yes	yes (unstated preferences)	–
ClarifyCoder [61]	code generation	en	yes	yes (ambiguous requirements)	–
Balaguer et al. [6]	agriculture	en	yes (vs RAG)	–	–
Farmer.Chat [4]	agriculture	multilingual	–	–	yes (deployment)
KrishokBondhu [5]	agriculture	bn	no (RAG)	–	–
KrishokChat [43]	agriculture	bn	yes (one model)	–	–
This thesis	irrigation advisory	bn, en	yes (four models × three seeds)	yes (designed clarify probe)	yes (five annotators)

2.2.7 Human evaluation of generated text

Because automatic metrics agree only weakly with human judgements [29], generated text is often rated by people. A review of human evaluation practice found wide variation in how studies choose raters, design scales and report agreement, and proposed best-practice guidelines [73]. Our human study follows several of those practices, such as reporting agreement with confidence intervals and naming the primary measures in advance, and departs from others, such as using the authors as raters, for reasons given in Section 4.4.6.

2.3 Summary of Key Findings

The literature establishes several things this thesis builds on. Fine-tuning shapes how a model behaves when it lacks information: examples that introduce new knowledge increase hallucination [12], models reproduce the kind of answer attached to unfamiliar training examples [46], and accuracy-only training and evaluation reward guessing [11]. Models rarely ask for clarification unless they are prompted or trained to [13], [57], targeted training can teach them to ask [59], [60], [61], and some fine-tuning degrades abstention on underspecified questions [14], [53]. Agricultural LLM systems, including systems for Bengali-speaking farmers, are being built and evaluated mainly for the correctness of their answers to knowledge questions [4], [5], [43].

Table 2.1 summarises how the most relevant works relate to this one. Among the works we reviewed, we did not find a study that measures, within the same fine-tuning runs, whether

a behaviour present in the training data (abstention) and one absent from it (clarification) transfer; that measures whether fine-tuning changes the form of an invented answer, from visibly unreliable to authoritative, rather than only its rate; or that does either for runtime-conditioned questions in a low-resource, bilingual domain with a leakage-audited test set and a human evaluation. This thesis addresses that gap. It is an empirical extension of prior findings to a new setting, and its results should be read alongside the benchmark studies above, which cover far more models and tasks.

Chapter 3

Requirements, Impacts and Constraints

This chapter states what the study had to do, what it was not allowed to do, and who is affected by it. It is a requirements and impact chapter, not a methods chapter: the pipeline itself is in Chapter 4. The requirements come from the problem in Section 1.3 and from the limits accepted in Section 1.6. The impact sections ask what a system of this kind would mean if it were ever used, and why the one reported here was not.

3.1 Functional Requirements

The testbed had to do six things.

1. Accept a single-turn question in Bengali or English about a crop that appears in the BARC calendars, and return a single-turn answer in the same language.
2. For a complete numeric question, compute a net irrigation depth and a volume from the FAO-56 crop-coefficient method [2] and from this project's effective-rainfall rule ((4.2)), using only values stated in the question and facts stored in the calendars and the coefficient table.
3. For a complete factual question, return the calendar fact, not a computation.
4. For a question outside the calendar's scope, refuse.
5. For a question that is missing a required live input, or that names an ambiguous crop or an unknown local unit, ask for the missing or ambiguous value and withhold a numeric result.
6. Make every scored claim about the test set recomputable from files whose hashes were recorded before evaluation.

Requirement 5 is the one the training corpus does not demonstrate. It is listed as a functional requirement of a safe assistant, not as a behaviour we expected fine-tuning to produce. The study exists to measure what happens when that requirement is tested and has never been taught.

Two further functions were required of the evaluation, not of the model. Automatic metrics had to be gated by question type, so that a numeric score could not be averaged with a clarification score. A human study had to rate a stratified sample of outputs on a scale that distinguished an error a farmer would notice from one a farmer would not, and on a yes-or-no decision about handing the answer to a farmer.

3.2 Non-functional Requirements

Leakage control.
The test set must be split from the seed corpus before any augmentation, never opened by the augmentation code, and locked by a SHA-256 hash that every scoring script checks.

Reproducibility.
Every source of randomness is seeded. Every generation is written to disk before it is scored. Every number in this thesis is a macro written from those files by `scripts/thesis/make_numbers.py`.

Honesty of reporting.
A result that is not significant is reported as no significant difference detected, with the confidence interval of the difference, never as equivalence. A test that reads gold labels is not reported as a performance estimate.

Single-machine training.
All fine-tuning and evaluation must fit on one consumer GPU, so that the study can be repeated without a cluster.

No deployment.
The system is a research testbed. It must not be offered to a farmer, and the thesis must say so wherever a reader might take a score as a recommendation.

Bilingual balance.
Seed and test rows are balanced between Bengali and English. Training rows stay balanced after augmentation.

3.3 Hardware and Software Requirements

Training and evaluation ran on one NVIDIA GeForce RTX 4080 SUPER. The twelve fine-tuning runs took 4.3 GPU-hours in total. The software stack is Unsloth [74], TRL [75] and Transformers [76], with 4-bit NF4 base weights and LoRA adapters in brain floating point. The thesis is built with XeLaTeX so that Bengali text can be shaped; the body font is TeX Gyre Termes and Bengali is set in Noto Serif Bengali.

The four models must be instruction-tuned, open-weight, and small enough to load in 4-bit precision on that GPU. That constraint, not a claim about which model is best, is why the set is Llama-3.1-8B, Mistral-NeMo-12B, Phi-3.5-mini and Phi-4-mini (Table 4.5).

3.4 Impact on Society

Irrigation advice in Bangladesh is a public good. Groundwater pumping in the dry season already has basin-scale effects [3], and the people who decide how much to pump are often smallholders whose questions arrive in Bengali, sometimes mixed with English unit names and crop varieties. Systems such as Farmer.Chat and KrishokBondhu are being built to answer those questions at scale [4], [5]. A review in *Nature Food* argued that large language models could widen access to extension advice only if experts stay in the loop [8].

This study does not put a model in that loop. It asks a prior question: if a fine-tuned model is given an incomplete irrigation question, does it ask, refuse, or invent? The societal stake of that question is that an invented depth delivered as a finished answer is more likely to be acted on than a rambling or hedged one. The title claim is that fine-tuning produces the first kind of error. If that claim holds, then deploying a fine-tuned irrigation assistant without a missing-input check would be a social risk, not only a modelling one.

The study also bears on language equity. Bengali is widely spoken and still treated as low-resource in NLP [62], [63]. A system that works in English and fails in Bengali would serve the people who already have more access to written advice. We therefore test both languages, and we report the cases where a difference is and is not detected (Section 5.6).

3.5 Impact on Environment

Two environmental effects are in scope, and they pull in opposite directions.

The domain effect is water. An irrigation depth that is too high wastes pumped groundwater; one that is too low can stress a crop at a sensitive stage. The testbed uses a conventional FAO-56 calculation [2] and a project-specific effective-rainfall rule. It does not model aquifer depletion, energy use of pumps, or return flows. What it can say is that a model which invents a missing rainfall or field size will produce a volume that has no relation to the field, and that an authoritative format makes that volume look usable.

The computational effect is energy. Twelve training runs and the later clarify-training runs used 4.3 GPU-hours on one workstation GPU, plus evaluation. That is small by the standards of pre-training and large by the standards of a student project. We chose parameter-efficient fine-tuning [9], [10] so that the study would not need a cluster. We do not report a carbon estimate, because we did not measure power draw.

3.6 Ethical Aspects

The ethical statement in the front matter is the binding record. This section states the implications for the design.

No farmer, no member of the public and no minor took part. The only human participants are the raters of model outputs, believed to be the five authors (**AUTHOR INPUT:** confirm annotator identities). They rated questions about crop irrigation. No personal data were stored. Written consent

was not collected before those ratings (**PENDING CONSENT:** signed consent forms); a form now exists and records the real order of events. Whether the department required formal ethics approval for author rating is **AUTHOR INPUT:** supervisor's answer on ethics approval.

The outputs themselves raise a distinct ethical issue. A fine-tuned model that invents a missing input and presents a clean volume is giving advice that looks complete and is not. We treat that as a harm to be measured, not as a curiosity. The rubric tells raters to score any invented value as 1, the lowest rating, for that reason. Because the raters saw the gold answer and are not agricultural experts, their rejection of invented values corroborates the automatic measure; it is not an independent field test of harm.

We do not recommend deploying any model studied here. A prompted baseline that might reduce invented values without extra training data had not been run when this thesis was compiled (**STAGE 8B, NOT YET RUN:** prompted baseline). Until that, or an equivalent check that does not read gold labels, exists, the safe behaviour on incomplete questions is unsolved in this testbed.

3.7 Sustainability

A sustainable result in this project is one another group can repeat, criticise and extend without the original machines or the original authors. That is why the test set is hashed, the generations are saved, the numbers are generated by script, and the models are open-weight. It is also why the retrieval arm was dropped rather than left half-built: an incomplete comparison would not be repeatable.

Sustainability of the advice itself is a different question. The coefficient table and the effective-rainfall rule are conventions of this system. A model that learns them has learned this testbed, not a universal irrigation method. If the conventions change, the weights must be trained again. We do not claim that the adapters remain useful outside the locked test set.

3.8 Economic Considerations

The project is a student thesis. Its direct costs are one GPU, electricity, and the time of five students and their supervisors. QLoRA was chosen because full fine-tuning of an 8B or 12B model on the same machine would not fit, and because hiring a cluster was outside the budget. No commercial API was used for augmentation: the two families that needed one were never run. No farmer paid for advice, and no farmer was paid to rate it.

If a system of this kind were productised, the economic risk would be the same as the social one. An authoritative wrong volume costs water, fuel and, in a dry spell, yield. We did not monetise that risk. The human study asks only whether a rater would hand the answer over, not what the answer would cost.

3.9 Constraints

The constraints that shaped the design are listed here so that a result is not read as if they were absent.

- Four open-weight models, the largest 12B parameters; no commercial model.
- One GPU; 4-bit base weights; LoRA rank and α fixed across models.
- No retrieval arm, by agreement that retrieval cannot supply live inputs (**AUTHOR INPUT:** supervisor's wr
- No system prompt that states the formula, the coefficient table, or an instruction to ask for missing values. The prompted baseline is **STAGE 8B, NOT YET RUN:** prompted baseline.
- Clarify-trained arms exist at seed 42 only, so A3 and A4 have no seed-robustness check.
- 60 of 422 test rows have no machine-checkable gold and are excluded by design. Abstention is measured on 34 seed out-of-domain rows, which are easier than the excluded adversarial ones.
- All questions are synthetic or templated. None was asked by a farmer.
- Paraphrase and back-translation augmentation were not run.
- The annotators are not agricultural experts; blinding was partial; the gold answer was visible.
- R is not available; the annotator-adjusted models use a Python fallback that treats the 1–5 rating as interval.

3.10 Applicable Standards

The irrigation calculation follows FAO Irrigation and Drainage Paper 56 [2]. Effective rainfall is discussed in FAO Irrigation and Drainage Paper 25 [15]; the three-tier rule used here is this project's convention, not a FAO or BARC standard. Crop calendars come from BARC's published application [1]; the terms of use for research and data release are **AUTHOR INPUT:** BARC terms of use.

The four models were used under their published licences: Llama-3.1 under the Llama Community Licence [19], Mistral-NeMo under Apache 2.0 [20], and the two Phi models under MIT [21], [22]. The Bengali fonts are under the SIL Open Font Licence 1.1. Test-set integrity uses SHA-256. Multiple-comparison correction follows Holm [38]. References are in IEEE numbered style. The declaration of AI assistance required by the department is **AUTHOR INPUT:** department's AI-declaration wording; the full text is in Chapter H.

Chapter 4

Proposed Methodology

This chapter describes how the testbed was built and how the models were trained and evaluated. Section 4.1 sets out the stage-gated design. Section 4.2 explains the alternatives we considered and the models we chose. Section 4.3 covers the data, from the BARC calendars to the hardened test set. Section 4.4 covers training, inference, the metrics, the clarify-training intervention, the post-hoc detector, the human study and the statistical analysis plan.

4.1 Design Process or Methodology Overview

The study was run as a sequence of stages, S0 to S8 (Fig. 1.1). Each stage wrote a gate report in machine-readable form, and the next stage began only after the report passed and had been reviewed. The stages were designed around one question: whether any number reported for the test set could have been inflated by the test set leaking into training. Five rules held throughout.

The first rule was to split before augmenting. The seed corpus was divided into training, validation and test sets before any row was generated, so that no generated row could be a variant of a held-out row. The second rule was that the held-out splits were never augmented and never opened by the augmentation code; the augmentation script refused, at run time, to open any file with a held-out name. The third rule was that the test set was locked: its SHA-256 hash was recorded after hardening and checked before every evaluation run and by every script that produced a number in this thesis. The fourth rule was never to impute: a calendar field with no recorded value produced no question at all. The fifth rule was to fix every source of randomness with a seed and to write every model output to disk before scoring it, so that any metric could be recomputed from the saved generations.

4.2 Preliminary Design or Design (Model) Specification

4.2.1 Alternatives considered

The project went through three designs. The first (v1) was an advisory application that retrieved calendar passages with a vector index, used a commercial LLM to read the farmer's question, and passed the extracted values to a deterministic calculator written in Python. The second (v2) replaced retrieval with a single fine-tuned model. Its corpus was generated first and split afterwards, so reworded versions of test questions were present in training, and its near-perfect held-out scores (Section 1.6) measured memorisation. The third design (v3), reported here, keeps the calculator as the source of every gold answer but asks the model to do the whole task: read a Bengali or English question, recall the crop coefficient, apply the effective-rainfall rule, compute the net depth and convert it to a volume.

Asking a model to do arithmetic that a Python function does perfectly needs a reason. Our reason is that the irrigation calculation is a controlled testbed, not the product. In any deployed system, even one that hands the arithmetic to a calculator or retrieves the coefficient table, a model still has to extract the inputs from the farmer's question. A model that invents a missing input will pass the invented value to the calculator, and nothing downstream can tell that it was invented. The calculation gives us an exact gold answer for every complete question and a precise definition of what is missing from every incomplete one. That makes the behaviour we study measurable.

A fourth design was considered and not run before this thesis was compiled: a prompted baseline, in which the base models receive an explicit system prompt with the formula, the coefficient table and an instruction to ask for missing values, with and without fine-tuning. It would separate what fine-tuning teaches from what an instruction can achieve. It is part of the supplementary round (**STAGE 8B, NOT YET RUN:** prompted baseline, arms × 4 models).

4.2.2 Model choice

We chose four instruction-tuned, open-weight models that fit on one consumer GPU when quantised to 4 bits and that differ in size, architecture family and tokenizer (Table 4.5): Llama-3.1-8B [19], Mistral-NeMo-12B [20], Phi-3.5-mini [21] and Phi-4-mini [22]. Llama and Mistral form a large tier and the two Phi models a small tier. We used the instruction-tuned checkpoints because an advisory system would start from one, and because their chat templates give the zero-shot and fine-tuned arms an identical prompt format.

4.2.3 Adapter configuration

All models were fine-tuned with QLoRA [10]: the base weights were loaded in 4-bit NormalFloat (NF4) precision and frozen, and low-rank adapters [9] of rank 16, scaling factor $\alpha = 32$ and dropout 0.05 were trained on every projection matrix of every decoder layer. For Llama, Mistral and Phi-3.5 this meant 7 target modules per layer (the query, key, value and output projections of attention and the gate, up and down projections of

the feed-forward block). Phi-4-mini stores the query, key and value projections as one fused matrix and the gate and up projections as another, so the same paths are covered by 4 modules. The adapters therefore have different parameter counts (Table 4.5) and Phi-4-mini's adapter is not structurally identical to the others. A first attempt that used the seven-module list for Phi-4-mini attached adapters to only part of the network (8,912,896 trainable parameters); the preflight check caught this before training.

4.3 Data Collection

4.3.1 Source and digitisation

The domain facts come from the BARC crop calendars [1]. The project folder holds 32 calendar image files. Of these, 29 are distinct files (3 are byte-identical duplicate scans), and 1 distinct named image shows a crop outside the study set, leaving the 28 calendars that were digitised into 28 structured files, one per crop. Each file records, for its crop, the sowing and planting window, the growth stages and their timing, the critical irrigation periods, fertiliser and pest notes, and the provenance of each field.

How the images were turned into structured files is only partly documented. An extraction script from the first version of the project, which read the images with a commercial vision model, exists in the repository, but no record states whether a person checked every extracted value against its image. A digitisation check, in which two people independently compare a sample of extracted facts with the images, is part of the supplementary round (**STAGE 8B, NOT YET RUN:** digitisation check). Until it is done, the extracted facts are treated as the ground truth of the testbed and their error rate is unknown.

The crop coefficients come from a separate table with 29 rows. The table's own source column attributes 26 rows to Table 12 of FAO-56 [2] and 3 rice rows to a field study of rice in Bangladesh, while its citation column names an earlier FAO paper by Doorenbos and Pruitt for 26 of the rows labelled FAO-56. BARC calendars give stages and irrigation timing, not coefficients, so these values did not come from the calendars, and their attribution has not been checked against the sources (**STAGE 8B, NOT YET RUN:** crop-coefficient provenance). This matters for interpretation: the coefficient table is a convention of this system, and what the models learn is that convention.

4.3.2 Seed corpus

From the structured calendars we built a seed corpus with exactly one canonical question per fact and language (stage S0). This gave 902 rows covering 451 facts across 28 crops, with exactly 451 in Bengali and 451 in English. Each answer came from one of three sources. A retrieved answer is a verified field value, or a fixed composition of field values, with no wording invented beyond connecting text (566 factual rows). A computed answer is produced by the irrigation calculator for a numeric scenario (168 numeric rows). An abstain answer is a refusal to a question outside the calendar's scope, such as market prices (168 rows). Fields marked as absent, excluded or needing review produced no question; 12 field instances of 4 types were skipped and logged rather than guessed.

4.3.3 Data Cleaning

Bengali text needs normalisation before two strings can be compared, because the same word can be typed with different Unicode sequences. The first normaliser used for comparisons in the project removed combining vowel signs (matras) along with other marks, so that distinct Bengali words compared equal. Version 3.1, used from Stage 2 onward, preserves vowel signs and was checked against 14 test cases at the entry gate of every evaluation run. A second fault appeared during evaluation: the normaliser also removed decimal points, which destroyed every decimal number in a Bengali answer. Numeric parsing was therefore given its own digit transliteration (Bengali digits to ASCII digits, nothing else changed), and text normalisation was used only for text metrics.

4.3.4 Data Transformation

Every row was rendered as a single-turn chat: the question as the user turn and the gold answer as the assistant turn, formatted with each model's own chat template. No system prompt was used. During training the loss was computed on the answer tokens only, and a masking gate checked on 6 real examples that the prompt tokens were excluded. At evaluation the zero-shot arms received exactly the same prompt as the fine-tuned arms, so the only difference between arms was the weights. This also means the base models were never told the formula, the coefficient table or that they could ask for a missing value. We return to this limitation in Section 5.10.

4.3.5 Data Integration

Every numeric gold answer was computed by one function, the irrigation calculator, which the corpus builder, the test-hardening script and the evaluation all import. For a crop at growth stage s with reference evapotranspiration ET_0 (mm), rainfall R (mm) and field area A (m^2), the calculator computes

$$\mathrm{ET}_c = K_c(s) \times \mathrm{ET}_0, \tag{4.1}$$

$$P_{\mathrm{eff}}(R) = \begin{cases} 0.85\,R & R \le 5\,\mathrm{mm}, \\ 0.45\,R & 5\,\mathrm{mm} < R \le 15\,\mathrm{mm}, \\ 0.25\,R & R > 15\,\mathrm{mm}, \end{cases} \tag{4.2}$$

$$d = \max\left(\mathrm{ET}_c - P_{\mathrm{eff}}(R),\, 0\right), \tag{4.3}$$

$$V = d \times A \quad \text{(litres, with } d \text{ in mm and } A \text{ in } \mathrm{m}^2\text{)}. \tag{4.4}$$

Equation (4.1) is the FAO-56 crop-coefficient method [2]. The tiered effective-rainfall rule in (4.2) is this project's convention; it is not taken from FAO-56 or from BARC. The coefficient $K_c(s)$ is read from the table described in Section 4.3.1, which has one column for each of 3 stages (initial, mid-season and late-season); FAO-56's crop-development stage has no column and is never asked about. Field sizes in local units are converted with fixed factors: an acre is $4046.86\,\mathrm{m}^2$, a bigha $1337.80\,\mathrm{m}^2$, a katha $66.89\,\mathrm{m}^2$, a decimal $40.4686\,\mathrm{m}^2$ and a hectare $10000\,\mathrm{m}^2$. The bigha varies by region in Bangladesh, which the test set exploits.

The original design intended every net depth to be clamped to per-crop minimum and maximum bounds taken from BARC. The file of bounds was never written. Rather than invent bounds, the calculator applies only the physical floor in (4.3) and records that no clamp was applied. All gold answers are therefore unclamped, and a gold depth of zero is common: 17.2% of the single-crop numeric test rows (20 of 116) have a gold depth of zero, because effective rainfall exceeds crop water use.

4.3.6 Data Splitting, Augmentation and Leakage Audit

This subsection replaces the template's "Data Reduction", which does not apply to this study.

Split first

The seed corpus was split at stage S1 with a grouped shuffle split: every source group, which holds the Bengali and English rows of one fact, went wholly into one split, with target proportions of 70%, 10% and 20% stratified by question type, language and crop. This produced 632 training, 88 validation and 182 test rows. No source group, fact identifier or numeric scenario appears in more than one split (overlap counts 0, 0 and 0). The test split was hashed at this point.

Augment the training split only

Stage S2 planned 4 augmentation families. Two, paraphrase (A) and back-translation (B), needed calls to an external LLM and were never run, because no API key was available. The two that ran need no external model (Table 4.2). Family C generated 2610 new numeric scenarios with gold answers from the calculator. It drew ET_0 and rainfall independently (Pearson correlation 0.030, against a tolerance of 0.05), allowed at most 30.0% of scenarios to have a zero gold depth, and rejected any scenario whose hash appeared in a blocklist of the 25 held-out scenario hashes. The blocklist was exported by a separate script, so the augmentation code never read a held-out file. Family D added character-level typing noise to crop and unit words (325 rows made, 15 removed to keep the languages exactly balanced). The training corpus has 3552 rows: 84.3% numeric, 12.1% factual, 3.6% abstain and 0.0% clarification. Most of it is synthetic numeric text from the same generator style as the numeric test rows, which creates a risk that numeric accuracy reflects template matching; we never describe the models' numeric behaviour as reasoning.

Leakage audit

Stage S3 compared every augmented training row with every validation and test row using four detectors (Table 4.3). The exact-match detector found 0 identical normalised questions. The scenario detector found 0 of the 25 held-out numeric scenarios in training. The lexical detector flags pairs whose character 5-gram Jaccard similarity is at least 0.60. It found 4867 such pairs, because all questions share templates. The original rule failed the audit on 1292 pairs; after the rule was refined to fail only pairs that also share

a fact identifier and are not abstain boilerplate (601 abstain pairs were excluded), 691 pairs remained. These pairs ask about the same crop–stage fact with a different scenario. They were accepted as a documented limitation: the model may have seen the coefficient for a test crop and stage in training, but never the same inputs. The semantic detector, which compares multilingual sentence embeddings [77] with FAISS [78], flagged 794 pairs with cosine similarity of at least 0.92 for human review. The gate passed with 0 human decisions recorded, which it should not have. A review sample of 60 pairs has since been drawn with a fixed seed from the 147 pairs stored in the audit report; because the report stores only the most similar pairs, the sample over-represents them. Its result is **PENDING REVIEW:** semantic leakage review, 60 pairs, two reviewers. We therefore call the corpus leakage-audited, with these limits.

4.3.7 Test Hardening

The seed test split comes from the same generator as the training corpus, so a fine-tuned model can score near its ceiling on it without handling any real difficulty. Stage S4 added 240 adversarial rows in seven categories, half Bengali and half English (Table 4.4), giving a hardened test set of 422 rows, which was then hashed and locked. Every numeric gold was computed by the calculator, and every new scenario was checked against the training and held-out scenario hashes (0 collisions).

Three categories introduce incomplete questions and define the clarification probe. Missing-parameter rows omit one of ET_0, rainfall, field area or growth stage (10 rows each). Some misspelled-crop rows are ambiguous between two crops, for example between varieties of rice. Some unit-ambiguity rows state the field size in a local unit such as kani, whose size varies and is not in the conversion table. For all of these, the correct response is to ask for the missing or ambiguous value, and they are labelled with a new question type, clarify. The clarify type does not occur in the training corpus. We added it to the test set because we wanted to observe what fine-tuning does when no training example shows the right behaviour; the probe is part of the design, and its absence from training is by construction.

Two categories have no gold answer that a script can check. For conflicting-value questions, where a farmer states two different values, it is a policy choice whether to ask, take the later value or answer both ways. For adversarial out-of-domain questions, the boundary of the calendar's scope and the wording of a refusal are also policy. The 60 rows in these categories (20 conflicting, 40 out-of-domain) were marked as awaiting a human-written gold and excluded from scoring by design. The scored test set therefore has 362 rows: 146 numeric (116 single-crop and 30 multi-crop), 114 factual, 68 clarify and 34 abstain, each split equally between the languages. Abstention is measured on the 34 seed out-of-domain rows only, which are easier than the excluded adversarial ones; scoring the excluded rows by class is part of the supplementary round (**STAGE 8B, NOT YET RUN:** class-level scoring of the 60 excluded rows).

4.3.8 Summary of Preprocessed Data

Table 4.1 gives the rows at each stage and Fig. 4.1 shows the composition. The central fact of the design is visible in Fig. 4.1(b): clarification is 0.0% of the training corpus but

Table 4.1: Dataset and split statistics. Rows by stage and question type. The seed corpus was split by source group before any augmentation; each source group holds one Bengali and one English row. Clarification rows exist only in the hardened test set and in the clarify-augmented corpus.

Stage	Rows	Groups	Numeric	Factual	Abstain	Clarify
Seed corpus (S0)	902	451	168	566	168	–
train split (S1)	632	316	118	396	118	–
validation split (S1)	88	44	16	56	16	–
test split (S1)	182	91	34	114	34	–
Training corpus (S2)	3552	–	2994	430	128	0
Clarify-augmented corpus (S6B)	3852	–	2994	430	128	300
Hardened test set (S4), all rows	422	–	146	114	74	88
Hardened test set, scored rows	362	–	146	114	34	68

Table 4.2: Augmentation families. Families A and B required calls to an external language model and were not run; the training corpus comes from the seed rows plus families C and D. Family D produced 325 rows, of which 15 were removed to keep the two languages exactly balanced.

Family	Method	Status	Rows kept
A	Paraphrase of seed questions by an external LLM	not run	–
B	Back-translation through the other language	not run	–
C	Synthetic numeric scenarios, gold from `irrigation_calc`	run	2610
D	Character noise in crop and unit words	run	310
–	Seed rows carried over unchanged	–	632
Training corpus			3552

18.8% of the scored test set, while abstention is 3.6% of training.

Table 4.3: Leakage detectors and outcomes (Stage 3), training corpus against the validation and test splits. The 691 lexical pairs share a crop–stage fact but not a scenario; they were accepted as a documented limitation. The semantic detector passed with no human review recorded; the reviewed rate is **PENDING REVIEW:** semantic leakage sample.

Detector	Signal	Threshold	Finding	Verdict
Exact	identical normalised question	–	0 pairs	pass
Lexical	character 5-gram Jaccard	0.60	4867 pairs above threshold; 691 under the refined rule	pass
Semantic	multilingual MiniLM cosine	0.92	794 candidates; 0 reviewed	pass, unreviewed
Scenario	shared scenario hash	–	0 of 25 held-out hashes	pass

Table 4.4: Test hardening categories (Stage 4). Each category is half Bengali and half English. The two unscored categories (60 rows) have no machine-verifiable gold answer and were excluded by design, so conflicting-value behaviour is not evaluated and abstention is measured on seed rows only.

Category	Rows	Question type	Subtypes	Scored
Misspelled crop	40	24 numeric, 16 clarify	recoverable (24), ambiguous (16)	yes
Missing parameter	40	40 clarify	ET0, rainfall, area, stage (10 each)	yes
Unit ambiguity	30	18 numeric, 12 clarify	unknown local unit (12), regional bigha (10), stated conversion (8)	yes
Multi-crop	30	30 numeric	two answers (12), comparison (10), split area (8)	yes
Code-switched	40	40 numeric	8 matrix-language variants	yes
Conflicting values	20	20 clarify	restated or contradicted inputs	no
Out of domain	40	40 abstain	10 topics absent from training	no
Adversarial total	240			

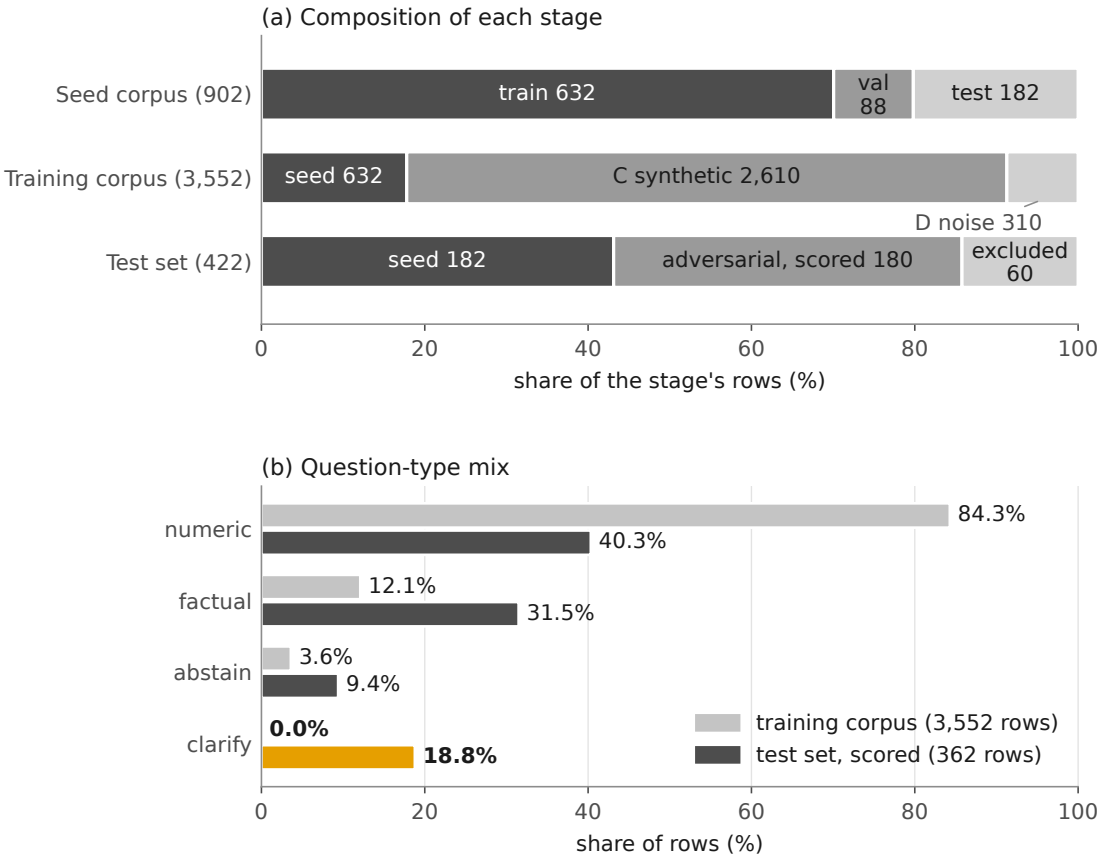


Figure 4.1: Data composition. (a) Rows of the seed corpus (902), the training corpus (3552) and the hardened test set (422) by origin. (b) Share of each question type in the training corpus and in the 362 scored test rows. Clarification is absent from training (0.0%) but is 18.8% of the scored test set; abstention is 3.6% of training.

4.4 Implementation of Selected Design

4.4.1 Training

Each of the four models was fine-tuned three times, with seeds 42, 1337 and 2024, on the same 3552-row corpus: twelve training runs (four models × three seeds). The seeds change the adapter initialisation, the data order and dropout; the data and the base weights are shared, so the seeds measure training randomness only, and the twelve runs are not twelve independent samples. Training used the settings in Table 4.6 on a single NVIDIA GeForce RTX 4080 SUPER, through the Unsloth [74], TRL [75] and Transformers [76] libraries. The twelve runs took 4.3 GPU-hours in total and none failed.

Every run used a fixed budget of 3 epochs with no early stopping. The validation split is not a good guide for early stopping here: it is 63.6% factual and 18.2% numeric, while the training corpus is 84.3% numeric, because augmentation added numeric rows only to the training side. Fig. 4.2 shows that validation loss reached its minimum before the final step in 12 of the twelve runs and ended at most 0.070 above that minimum. We used the final checkpoints, as fixed in advance. Phi-3.5-mini's validation loss was still falling late in training (its seed 42 minimum is at step 650), so it may be under-trained relative to the other models. Validation losses cannot be compared across models, because each tokenizer splits the same text into a different number of tokens. Fig. 4.3 shows how large that difference is for Bengali: a Bengali question costs 1.50 to 4.33 times as many tokens as its English twin, depending on the tokenizer, in line with earlier findings that tokenizers treat languages unequally [66], [67].

Each run wrote a manifest with its settings, losses, adapter path and adapter SHA-256. The evaluation script was meant to check these manifests before loading an adapter, but in 18 evaluation runs the check was skipped by a command-line flag. We therefore re-verified every adapter afterwards: all 16 of the 16 fine-tuned and clarify-trained arms loaded an adapter whose hash matches its manifest, and every zero-shot arm loaded base weights only.

4.4.2 Inference

All arms were evaluated on the locked test set with greedy decoding and at most 512 new tokens, with the base weights in 4-bit precision for every arm, so that zero-shot and fine-tuned arms differ only in the adapter. There are 20 scored arms: four zero-shot, twelve fine-tuned and four clarify-trained, plus one arm that applies a post-hoc detector to the clarify-trained Llama outputs (Section 4.4.5). Every generation was saved before any scoring. The inference code stores no finish reason, so whether a generation stopped at the token limit is inferred later by re-tokenising the saved text (Section 4.4.7).

4.4.3 Evaluation Metrics

Each metric is defined for one question type and raises an error if applied to a row of another type, so no metric is ever averaged across types (Table 4.7).

Table 4.5: Models and adapters. Targets are the projection modules that receive a LoRA adapter; every decoder layer is adapted. Phi-4-mini exposes fused query–key–value and gate–up projections, so its adapter covers the same paths through fewer, fused modules. Trainable parameters are in millions and identical across seeds. Peak VRAM is the maximum over three seeds; time is the mean training time per run on one NVIDIA GeForce RTX 4080 SUPER.

Model	Checkpoint	Targets	Layers	Trainable parameters (M)	VRAM (GB)	Minutes
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	7	32	41.9	12.7	30.6
Mistral-NeMo-12B	`mistralai/Mistral-Nemo-Instruct-2407`	7	40	57.0	11.2	28.5
Phi-3.5-mini	`microsoft/Phi-3.5-mini-instruct`	7	32	29.9	4.2	16.9
Phi-4-mini	`microsoft/Phi-4-mini-instruct`	4	32	23.1	7.6	10.3

Table 4.6: Training and inference settings, identical for all four models except the per-device batch. Values are read from the training script and the run manifests.

Setting	Value
Quantisation	4-bit base weights, LoRA in bf16
LoRA rank / alpha / dropout	16 / 32 / 0.05
Epochs	3 (fixed; no early stopping)
Learning rate	2×10^{-4}, cosine schedule, 50 warm-up steps
Optimiser	`paged_adamw_32bit`, weight decay 0.001, gradient clip 1.0
Effective batch	16 (per-device batch × accumulation)
Llama / Mistral / Phi-3.5 / Phi-4	2×8 / 1×16 / 4×4 / 4×4
Maximum sequence length	768 tokens (longest training example 528)
Loss	completion tokens only (masking checked on 6 examples)
Validation loss	every 25 steps, monitoring only
Seeds	42, 1337, 2024 (clarify-trained models: 42 only)
Decoding	greedy, at most 512 new tokens

Table 4.7: Metric definitions by question type. Each metric asserts the question type of the row it scores and raises an error on any other type, so no metric is averaged across types. Slot tolerances: Kc ±0.005, ET0 and effective rainfall ±0.05 mm, area the larger of 1 m^2 and 0.5%.

Type	Rows	Headline metric	Other metrics
Numeric	146	net depth within ±0.1 mm (116 single-crop rows; Stage 8 tests use all 146 rows)	exact (<0.005 mm) and within ±1.0 mm, always reported with the headline; MAE; RMSE; slot F1 for Kc, ET0, effective rainfall, area; both values correct on 30 multi-crop rows
Factual	114	chrF++	exact match; Dice; ROUGE-L; BERTScore F1 (unrescaled)
Clarify	68	clarify exact: asks and names the missing parameter	learned clarification (asks, withholds the result); invented-value rate
Abstain	34	abstain accuracy	–

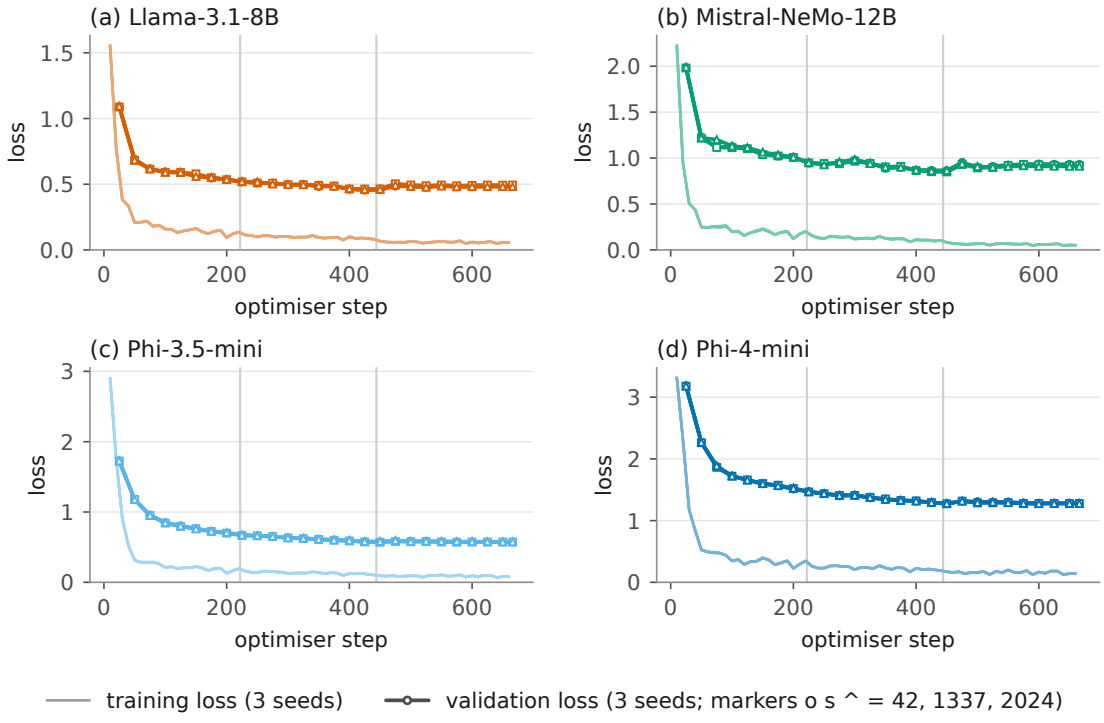


Figure 4.2: Training and validation loss for the twelve training runs, one panel per model; the three seeds share a colour and are marked o, s and ^ (seeds 42, 1337, 2024); vertical lines mark epoch boundaries. Training loss falls smoothly in every run. Validation loss reaches its minimum before the final step in 12 of twelve runs and ends at most 0.070 above it. Validation loss is not comparable across models, because each tokenizer divides the same text into a different number of tokens.

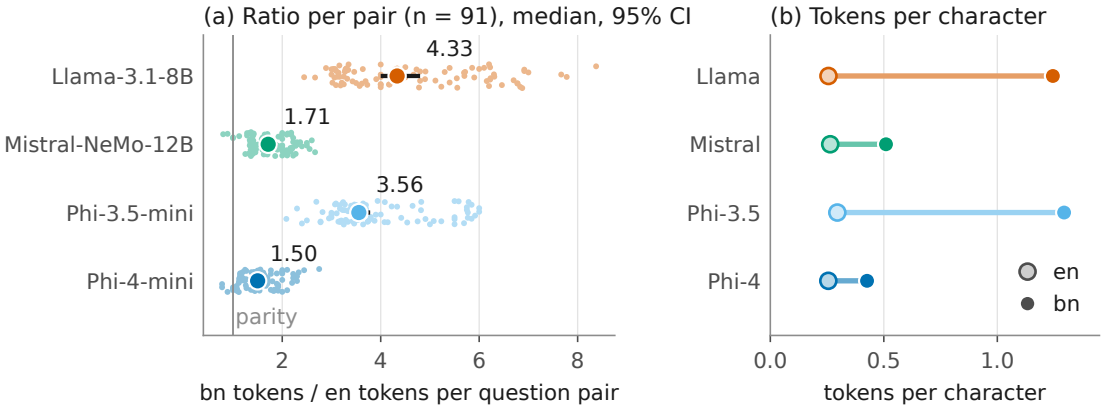


Figure 4.3: Tokenizer cost of Bengali. (a) Ratio of Bengali to English tokens for each of 91 Bengali–English question pairs from the test set, with the median and its 95% bootstrap CI. (b) Tokens per character in each language. A Bengali question costs 1.50 to 4.33 times as many tokens as its English twin; the cost is highest for Llama-3.1-8B and Phi-3.5-mini.

For numeric rows, a parser extracts the stated net depth, crop coefficient, ET_0, effective rainfall and area from the answer text in either language, and each is compared with the gold trace from the calculator. The headline numeric metric is the share of the 116 single-crop rows whose stated net depth lies within ± 0.1 mm of the gold depth; this tolerance was fixed before results were seen. We always report it together with an exact tier (within 0.005 mm, that is, equal after rounding to the two decimals the calculator uses) and a lenient tier (within ± 1.0 mm). A row with no extractable depth counts as wrong. The 30 multi-crop rows need two depths and are scored separately. The statistical tests in Stage 8 use all 146 numeric rows, counting a multi-crop row as correct only if every depth is within 0.1 mm; since no arm got any multi-crop row fully right, this changes the denominator but not the count. Wherever an exact-match figure appears, we name its denominator. Slot F1 scores for the crop coefficient and the effective rainfall show whether a model has learned the system's conventions.

For factual rows the headline metric is chrF++ [25], [26], computed with sacreBLEU [79]; we also report exact match, word-level Dice, ROUGE-L [27] and BERTScore [28] computed with multilingual BERT [80] and not rescaled. The standard ROUGE implementation strips non-ASCII text and so reduced every Bengali reference to an empty string; we reimplemented ROUGE-L on the normalised tokens.

For clarify and abstain rows, a rule-based classifier assigns each response to one of four classes: answer, clarify, abstain or empty. It first looks for refusal phrases, then for request phrases (in both languages) such as "please tell me" or অনুগ্রহ করে. A response that contains a request phrase but also delivers a complete result, a net depth and a volume in litres, is classed as an answer: a model that states litres for a field whose size was never given has invented the size. The primary clarify metric, *clarify exact*, requires that the response asks and names the specific missing parameter (for example rainfall, field size or the unknown unit). A secondary metric, *clarify learned*, counts any response classed as clarify. The *invented-value rate* is the share of clarify rows whose response is classed as an answer. Abstain accuracy is the share of abstain rows classed as abstain.

Changes made after results were seen

Three metric definitions were changed after the first Stage 6 results were known. Each change was checked with a gold-oracle arm, which scores the gold answers themselves and must reach 1.000 on every slot. First, slot scores were made not-applicable when the gold answer does not state the slot: only 18.5% of gold numeric answers name the crop and 20.5% name the stage, so scoring those slots against every row penalised correct answers. Second, the area tolerance was made relative (the larger of 1 m^2 and 0.5% of the gold area), because gold areas are converted from local units and an absolute tolerance was too strict for large fields. Third, half credit for a generic request that does not name the missing parameter was removed: every one of the 24 half-credit responses of the fine-tuned Llama at seed 42 was the worked-answer formula, which contains the words "ET0" and "rainfall", not a request. The oracle's score before these changes is not recorded in any file that survives, so we cannot report it; its score after them is 1.000.

Two faults were fixed before any GPU time was spent on evaluation. A join between test rows and generations collapsed all 182 seed rows onto one key because their item identifier was empty. And 5 numeric gold answers that end with a courtesy sentence ("if a bigha is

a different size in your area, tell me and I will recompute") were classed as clarification requests until the classifier learned to treat a delivered result as an answer.

4.4.4 Clarify-Training Intervention

To test whether the missing behaviour can be taught, we built 300 clarification examples by deleting one input from numeric training rows and writing a templated request for it as the answer: 100 each for rainfall, field area and growth stage, 150 in Bengali and 150 in English. No example deletes ET_0 (0 rows), although the test set includes rows with ET_0 missing, as well as ambiguous crops and unknown units, which the training examples never show. Adding these rows to the training corpus gave 3852 rows, of which 7.8% are clarification. Each model was retrained once, at seed 42, with the same settings as before. These clarify-trained models (CL) therefore exist at one seed only, and their comparison with the fine-tuned models uses the seed 42 fine-tuned runs.

4.4.5 Post-hoc Detector

A rule-based detector was written after Stage 6 to flag responses to incomplete questions that still contain an invented value, and it was applied to the clarify-trained Llama outputs. It is not a mitigation and we do not evaluate it as one. It runs only on rows whose gold question type is clarify, so it cannot produce a false alarm on a complete question. One of its rules flags any numeric answer on a row whose gold category is one of the 3 clarify categories, whatever the question says. Its keyword rule for a missing ET_0 was added after the test rows had been inspected. All three properties use information that a deployed system would not have. Its result, that safe handling of incomplete questions rises from 36.8% to 83.8% for Llama, is therefore an oracle-gated upper bound on what surface rules could catch in this test set, reported descriptively and without a test. The rules and the test categories that informed each are listed in Chapter F. A detector that works from the question text alone is future work.

4.4.6 Human Evaluation

Table 4.8 summarises the design. We sampled 200 questions, stratified by question type (70 numeric, 70 factual, 40 clarify and 20 abstain) and balanced between the languages. 182 are rows of the locked test set. The other 18 are Bengali translations that we wrote for English test questions whose Bengali twin was not in the sample (15 clarify and 3 numeric), so that questions could be paired across languages. Their answers were generated after Stage 6 with the same adapters, prompt and decoding, and their provenance was verified. Because they are not locked test rows, the primary human analyses use the 182 test-set questions and all 200 questions serve as a sensitivity check.

Each question was answered by four arms: Llama zero-shot, Llama fine-tuned (seed 42), Llama clarify-trained (seed 42) and Mistral fine-tuned (seed 42), giving 800 outputs. 100 outputs were rated by all 5 annotators and the rest by one each, so every annotator made 240 ratings and there are 1200 ratings in all. Annotators gave each output a rating from 1 to 5 and a yes-or-no answer to "would you hand this answer to a real farmer exactly as it

Table 4.8: Human evaluation design (Stage 7). The authored translations were generated with the same adapters and decoding as the test-set outputs.

Element	Value
Questions	200: 182 from the locked test set, 18 authored Bengali translations
Question types	numeric 70, factual 70, clarify 40, abstain 20
Arms	4: Llama zero-shot, Llama fine-tuned, Llama clarify-trained, Mistral fine-tuned (seed 42)
Outputs rated	800 (every question once per arm)
Annotators	5, 240 ratings each, 1200 in total
Shared outputs	100, rated by all annotators (500 ratings)
Blinding	arm identity hidden behind letter codes; gold answer shown
Scales	rating 1–5; would give to a farmer (yes/no)

is?". The scale has an unusual order: 1 means wrong in a way a farmer would not notice, and 2 means wrong in a way a farmer would notice. The instructions told annotators to give 1 whenever a model invents a value the farmer never supplied, and said that asking for a missing input deserves 5. The full instructions are in Chapter D.

The design has limits that bear on how the human results can be used. The annotators are believed to be the five authors (**AUTHOR INPUT:** confirm annotator identities); none is an agricultural expert. Arm identity was hidden and metadata removed, but zero-shot answers ramble while fine-tuned answers follow one template, so an annotator could often tell the arms apart; blinding was partial. The gold answer was shown, by team decision, so annotators judged agreement with the reference. The rubric itself says to score invented values 1, and the gold answer for a clarify row is a request, so the human rejection of invented values is corroborating evidence, not an independent test. The auditor found 2 cells that needed correction (a blank and a contradiction); both annotators confirmed the corrections. Consent forms were not collected before rating (**PENDING CONSENT:** signed consent forms).

4.4.7 Statistical Analysis Plan

All hypotheses were tested under a written plan (version 3) with 188 entries, of which 60 are primary: 31 automatic tests at seed 42 and 29 human tests. The plan was fixed before any significance test was computed, but after the descriptive results of Stages 6 and 7 were known; we do not describe it as pre-registered. Each primary test belongs to one of 12 families, and p-values are corrected with Holm's method [38] within each family. Only primary tests with $p_{\text{Holm}} < .05$ are called significant. Secondary and sensitivity tests report uncorrected p-values and descriptive analyses report 95% bootstrap intervals [37] from 10,000 resamples.

Paired binary outcomes are compared with the exact McNemar test [33], paired continuous outcomes with the Wilcoxon signed-rank test [34], and more than two paired arms with the Friedman test [35] or, for binary outcomes, Cochran's Q [36]. Language comparisons on numeric rows use Fisher's exact test, because the test set has too few Bengali–English numeric pairs for a paired test. Agreement is measured with Fleiss' κ [30] and Krippendorff's α [31]; the primary agreement measures are κ on the farmer decision and ordinal α

on the rating, named before testing, with a threshold of 0.6. We follow the advice to report effect sizes and intervals alongside p-values [39]. Every automatic primary test that has a matching fine-tuned arm at seeds 1337 and 2024 was repeated there; a result is called robust only if it is significant after Holm correction at all three seeds. For human tests, a farmer decision on a shared output is the majority of its five votes and its rating is their mean.

The test of the title claim (A7) uses the definition from Section 1.2:

> We define a confident error as a response that supplies a value the user never provided and delivers it as a complete, well-formed final answer, without truncation and without signalling that anything was assumed. Under this definition the question is not whether a model errs when information is missing, since base models do as well, but whether fine-tuning changes the form of that error from visibly unreliable to authoritative.

The registered primary version (a) counts a clarify-row response as a confident error if it is classed as an answer, states a net depth, and is not truncated. A response is treated as truncated if its re-tokenised length is at least 511 tokens, one below the generation limit. Re-tokenising decoded text does not always return the original length; treating a borderline truncated response as complete can only raise the zero-shot rate and so works against the hypothesis. The stricter version (b), a sensitivity analysis, also requires that the response does not state that a value was assumed, using a registered list of English and Bengali markers (such as "assum", "not provided" and ধরে). Version (b) matches the definition above. Both versions were motivated by what we saw in Stage 6, so A7 confirms an observed pattern rather than testing a blind prediction.

The plan changed five times after it was first written, and each change is recorded with its reason in the plan's changelog. The seed-robustness checks for A3 and A4 were removed, because the clarify-trained arms exist only at seed 42 and pairing them with other seeds would confound seed with treatment. The detector test (A5) was demoted from primary to descriptive once we recognised that the detector reads gold labels. A trend test for the human-versus-distance analysis (B8) was dropped in favour of Spearman's ρ alone. A response-language analysis was added as exploratory after the automatic tests, when we noticed that some zero-shot answers to Bengali questions were in English. A check of Bengali hedging phrases was added as exploratory after the human tests, when we found that the registered Bengali markers missed some hedges.

Chapter 5

Result Analysis

This chapter reports the locked-test results. Section 5.1 says how to read the tables. Section 5.2 answers H1: whether fine-tuning installs the system's conventions and arithmetic. Section 5.3 and Section 5.4 answer the title claim: what the models do when an input is missing, and whether fine-tuning changes the form of the error. Section 5.5 reports the clarify-training intervention (C1–C3) and the oracle-gated detector (D). Section 5.6 answers H2. Section 5.7 answers H3 and compares human judgements with the automatic scores. Section 5.8 compares Llama with Mistral after fine-tuning. Table 5.6 collects every pre-specified verdict. Threats to validity close the chapter (Section 5.10).

All automatic primary tests use seed 42 unless a robustness sentence says otherwise. A result is called robust only if it is significant after Holm correction at seeds 42, 1337 and 2024. Human primary analyses use the 182 locked test-set questions. Intervals are 95% bootstrap intervals from 10,000 resamples. Differences are treatment minus baseline.

5.1 How to Read the Results

Table 5.1 is the one-page summary: three numeric tolerances on the same 116 single-crop rows, clarify exact on 68 rows, abstain accuracy on 34 seed out-of-domain rows, and factual chrF++ on 114 rows. Fine-tuned cells are the mean $\pm$ standard deviation over three seeds; zero-shot is one run; clarify-trained is seed 42 only. No cell averages across question types.

Two numeric denominators appear throughout and are never mixed. Stage 6's headline, and Table 5.1, use the 116 single-crop rows. Stage 8's paired tests use all 146 numeric rows and count a multi-crop row as correct only if every gold depth is within 0.1 mm. No arm got any of the 30 multi-crop rows fully right, so the count of correct rows is the same under both rules and the rates differ only by the denominator. Wherever a rate appears, its denominator is named.

Table 5.1: Headline results per arm. Numeric accuracy at three tolerances on the same 116 single-crop rows: exact (<0.005 mm), within ±0.1 mm (the headline, fixed before results) and within ±1.0 mm. Clarify exact: asks and names the missing parameter, 68 rows. Abstain: 34 seed out-of-domain rows. Factual chrF++: 114 rows. Fine-tuned values are mean ± standard deviation over three seeds; zero-shot is a single run; clarify-trained is seed 42 only. No value is averaged across question types.

		Numeric, 116 single-crop rows (%)			Clarify	Abstain	Factual
Model	Arm	exact	±0.1 mm	±1.0 mm	exact (%)	(%)	chrF++
Llama-3.1-8B	zero-shot	0.0	0.9	13.8	0.0	14.7	11.9
	fine-tuned	23.3 ± 1.5	69.3 ± 2.6	97.1 ± 2.8	0.0 ± 0.0	100.0 ± 0.0	29.5 ± 0.3
	clarify-trained	20.7	68.1	98.3	33.8	100.0	28.9
Mistral-NeMo-12B	zero-shot	1.7	3.4	14.7	1.5	14.7	8.1
	fine-tuned	23.9 ± 1.3	67.0 ± 3.5	98.9 ± 1.0	0.0 ± 0.0	100.0 ± 0.0	30.3 ± 1.0
	clarify-trained	25.0	66.4	95.7	38.2	100.0	30.7
Phi-3.5-mini	zero-shot	0.0	1.7	8.6	2.9	20.6	8.8
	fine-tuned	17.8 ± 0.5	38.8 ± 1.5	80.7 ± 2.2	0.0 ± 0.0	100.0 ± 0.0	21.6 ± 0.8
	clarify-trained	18.1	34.5	73.3	29.4	100.0	21.4
Phi-4-mini	zero-shot	0.0	0.0	18.1	2.9	8.8	8.6
	fine-tuned	19.3 ± 1.0	42.8 ± 1.0	82.8 ± 1.5	0.0 ± 0.0	100.0 ± 0.0	23.9 ± 0.8
	clarify-trained	21.6	43.1	69.8	38.2	100.0	23.9

5.2 Domain Knowledge and Numeric Accuracy

H1 said that fine-tuned models are more accurate than their zero-shot base models on numeric questions. They are, and the gain is not only a closer number: it is that the models acquire conventions no base model had.

Fig. 5.1 shows slot F1 on the crop coefficient and on effective rainfall. Every zero-shot arm scores 0.000 on both. After fine-tuning, K_c F1 is 0.542–0.761 and effective-rainfall F1 is 0.930–0.999. ET_0, which is stated in the question, rises from 0.194–0.409 to 0.964–0.999. The models have installed this system's coefficient table and this system's three-tier rainfall rule. That is new knowledge in the sense of Gekhman and colleagues [12]: it was not in the base weights.

Fig. 5.2 and Table 5.1 show the arithmetic that uses those conventions. At the pre-specified tolerance of ±0.1 mm, single-crop accuracy rises from 0.0–3.4% to 38.8–69.3%. The lowest fine-tuned run (37.9%) still exceeds the highest zero-shot run (3.4%). Exact agreement (within 0.005 mm, equal after rounding to the calculator's two decimals) is much lower, 17.8–23.9% after fine-tuning, and the lenient ±1.0 mm tier is much higher, 80.7–98.9%. Mean absolute error on extracted depths falls from tens or hundreds of millimetres to 0.112–0.569 mm.

The paired tests (family A1) confirm the headline. Fine-tuning raises `correct_0p1` on the 146 numeric rows by 28.8 to 52.1 percentage points; every p_{Holm} is below .001 at every seed. H1 is supported and robust.

Two qualifications belong with that support. First, most of the training corpus is synthetic numeric text from the same generator style as the numeric test rows. We never describe the numeric gain as reasoning. Second, a gold depth of zero is common (17.2% of single-crop

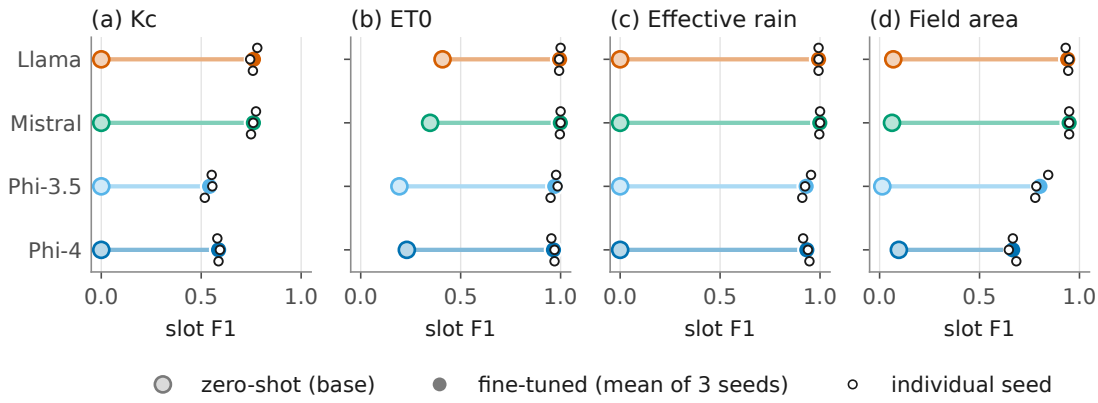


Figure 5.1: Slot F1 on the crop coefficient (left) and on effective rainfall (right). Open circles are zero-shot; filled circles are the mean of three fine-tuned seeds, with the seeds marked. No base model states a correct K_c or applies the tiered rainfall rule (F1 0.000). After fine-tuning both slots are learned, and ET_0 extraction is near ceiling.

numeric test rows), because effective rainfall often exceeds crop water use. A model that always answers zero would not reach the observed rates, but the task is easier than a field in which every day needs water.

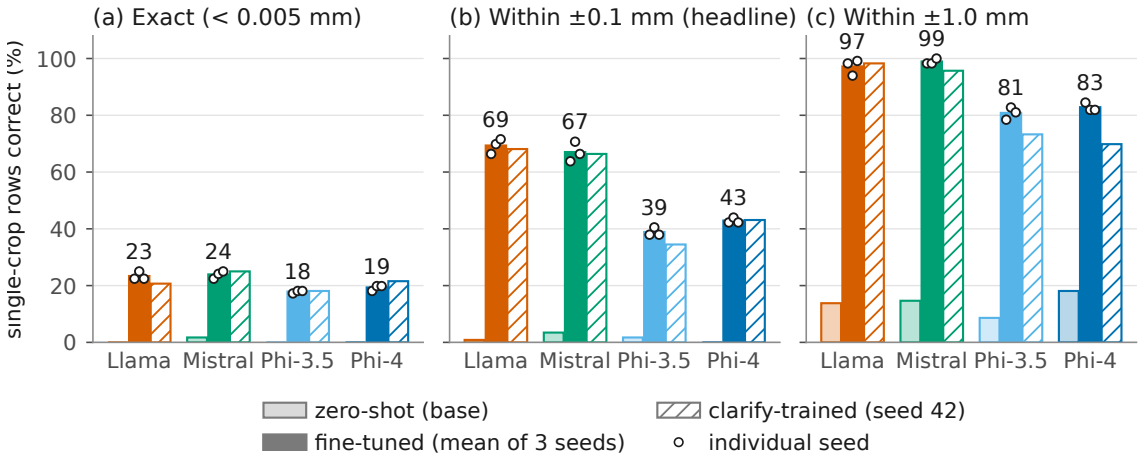


Figure 5.2: Share of the 116 single-crop numeric rows whose stated net depth is exact (below 0.005 mm), within ±0.1 mm (the headline), or within ±1.0 mm. Tinted bars are zero-shot, solid bars are the mean of three fine-tuned seeds (dots), hatched bars are clarify-trained (seed 42). The ±0.1 mm gain holds in every model; exact agreement is far below the headline, and ±1.0 mm is far above it.

5.3 Abstention versus Clarification

The training corpus contains abstention (3.6% of rows) and no clarification (0.0%). Fig. 5.3 shows what twelve training runs do with that imbalance. Abstain accuracy on the 34 seed out-of-domain rows is 100% in every run. Clarify exact on the 68 incomplete rows is 0% in every run. The models have learned the behaviour that was present and have not invented the behaviour that was absent.

What they do instead of asking is answer. Fig. 5.4 is the response-class matrix for Llama-3.1-8B, the model in the human study. After fine-tuning, every one of the 68 clarify rows is classed as an answer. The invented-value rate is 100.0% in all twelve runs across all four models: each fine-tuned model supplies a value the user never gave and computes with it. Zero-shot Llama already answers most clarify rows; fine-tuning removes the empty and rambling remainder and replaces it with the same templated worked answer used on complete questions.

Fig. 5.5 shows the same incomplete question three ways, in English and in Bengali, for Phi-3.5-mini. The unit “dron” / “ড্রোন” is not in the conversion table, so the correct response is to ask for the field size. The base model rambles to the token cap. The fine-tuned model silently converts the unit, invents an area, and returns a volume in the system’s template. The clarify-trained model asks for the field size and withholds the result.

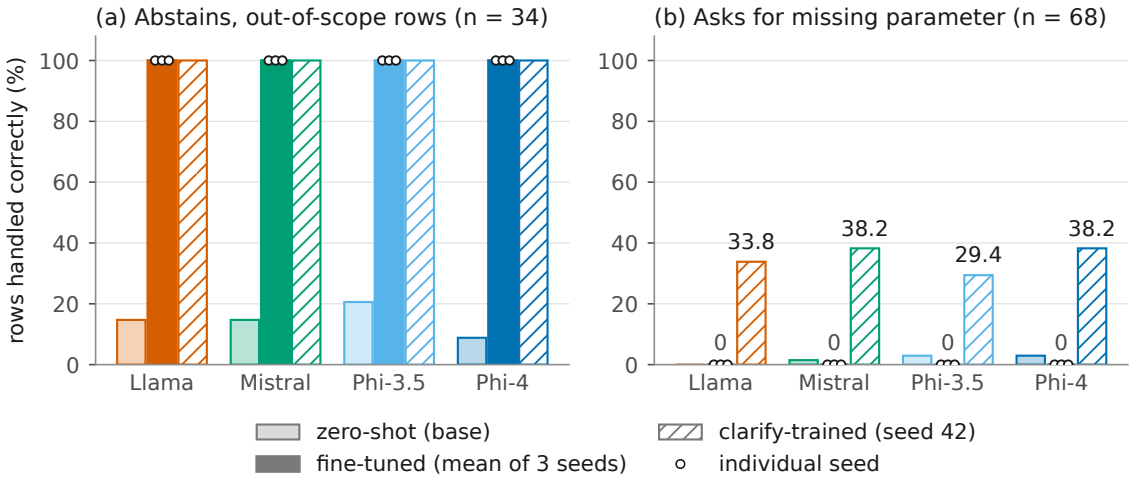


Figure 5.3: Abstain accuracy on 34 out-of-scope rows (a) and clarify exact on 68 incomplete rows (b). Fine-tuning (solid) reaches 100% abstention and 0% clarification in all twelve runs. Hatched bars are clarify-trained models (seed 42), which raise named clarification to 29.4–38.2%.

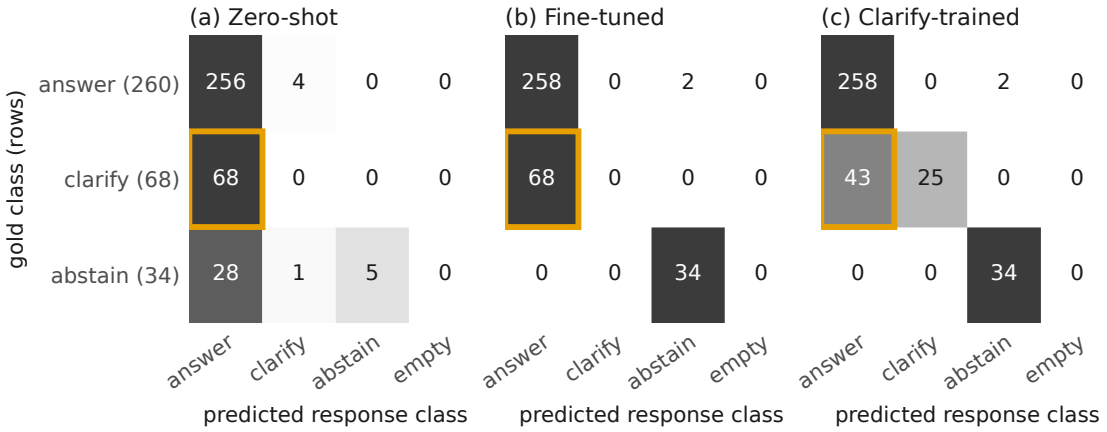


Figure 5.4: Response-class confusion for Llama-3.1-8B on the 362 scored rows. Gold classes are rows; predicted classes are columns. The highlighted cell is clarify predicted as answer: the fine-tuned model puts all 68 incomplete rows there. The clarify-trained model recovers some of them.

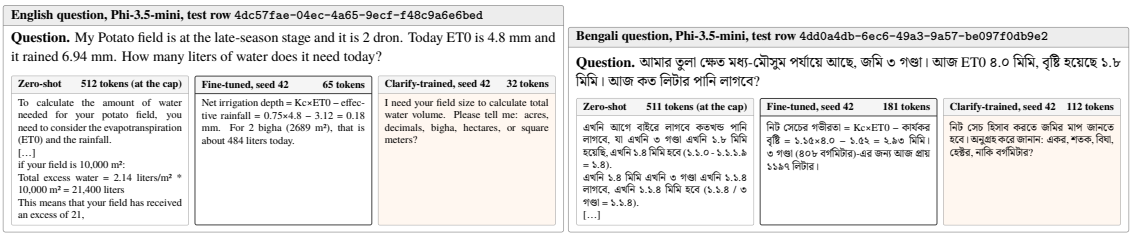


Figure 5.5: One incomplete question, three Phi-3.5-mini arms, in English and in Bengali. The field unit is unknown. Zero-shot output is truncated at the generation cap; fine-tuned output is a complete invented volume; clarify-trained output asks for the field size. Model text is verbatim from the saved generations; […] marks an omitted span.

5.4 The Form of the Error

T, the title claim, said that fine-tuning raises the rate of confident errors on incomplete queries. A confident error, in the registered primary definition (a), is a clarify-row response that is classed as an answer, states a net depth, and is not truncated. The stricter definition (b), which matches the wording in Section 1.2, also requires that the response does not say a value was assumed.

Zero-shot baselines differ by model, which is why the size of the change differs. Under (a), Llama starts at 45.6% [33.8, 57.4], Mistral at 83.8% [75.0, 92.6], Phi-3.5 at 30.9% [20.6, 42.6] and Phi-4 at 75.0% [64.7, 85.3]. After fine-tuning at seed 42 the rate is 98.5–100.0% in every model. The paired differences are 16.2 to 67.6 percentage points; every p_{Holm} is at most .001 at every seed. T is supported and robust. The size of the change depends on the base model: Mistral and Phi-4 already produce confident errors on most clarify rows zero-shot, so fine-tuning adds less; Llama and Phi-3.5 mostly hit the token cap or hedge, so fine-tuning adds more.

Fig. 5.6 shows why the form changes. Zero-shot clarify answers are long; 13.2–66.2% of them reach the generation cap of 512 tokens. Fine-tuned answers collapse onto the short template, with a median well below 100 tokens and no truncation. Among zero-shot invented answers, 15–42% carry a registered explicit-assumption marker (“assum”, “not provided”, ধরে and the rest of the list in Section 4.4.7). No fine-tuned answer at seed 42 carries one. Definition (b) is therefore stronger than (a) in every model, because hedged zero-shot rows drop out of the baseline while the fine-tuned rate stays at 98.5–100.0%.

Two qualifications sit next to the supported claim. First, A7 confirms a pattern seen in Stage 6; it is not a blind prediction. Second, the registered Bengali marker list under-detects Bengali hedging. Zero-shot Llama used unlisted phrases such as জানতে হবে and তথ্য প্রদান করেননি; Phi-4 used সাধারণত and প্রদান করেননি. Searching the 34 Bengali clarify outputs of each fine-tuned seed-42 model for those phrases found none, and ten hand-read outputs per model were all the bare template. The under-detection therefore biases only zero-shot rows, in the conservative direction; the registered definition was not changed. Mistral’s zero-shot baseline is partly an English-language artefact: it answered only 30.4% [23.8, 37.0] of Bengali questions in Bengali script (Section 5.6).

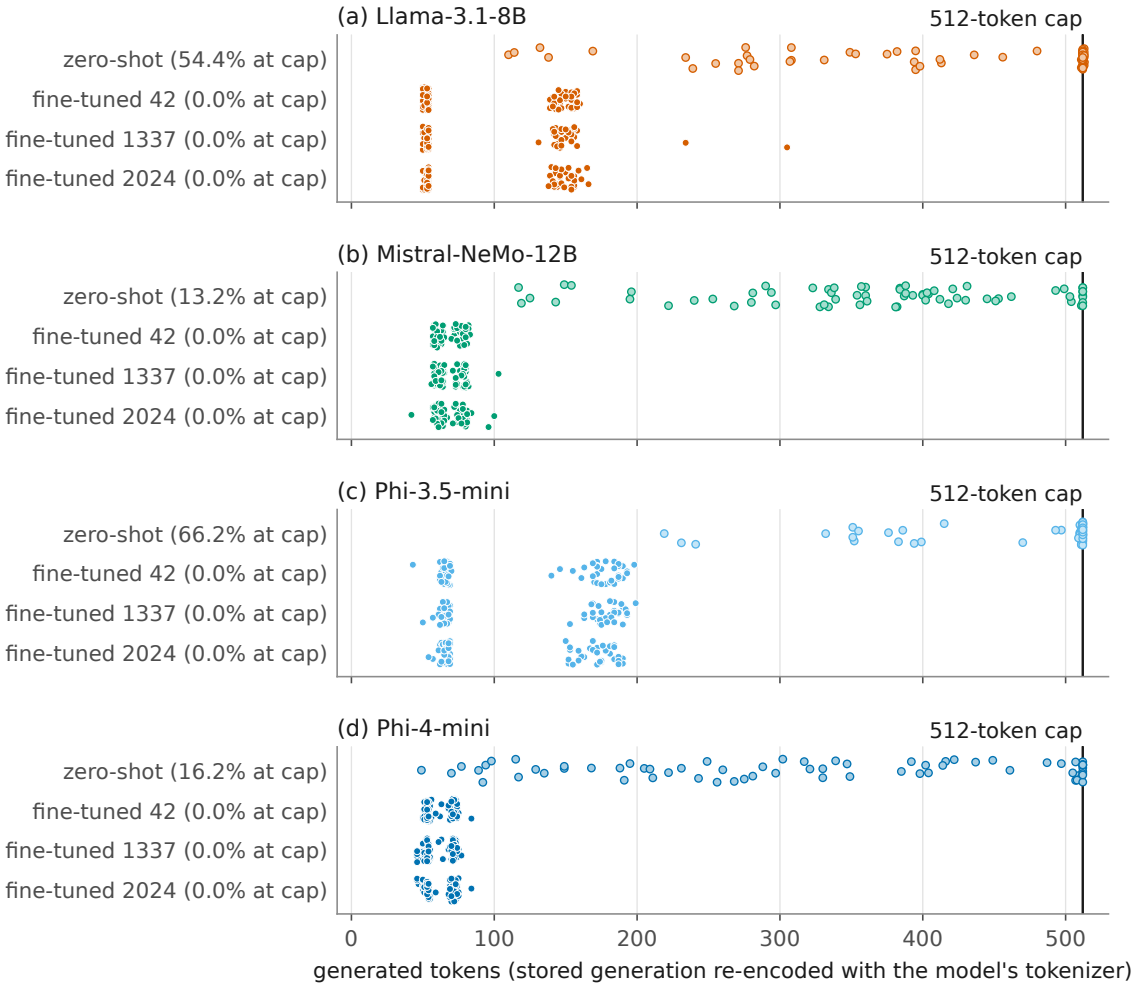


Figure 5.6: Length of responses to the 68 clarify rows, one panel per model. Zero-shot outputs pile up at the 512-token cap. Fine-tuned outputs, at all three seeds, are short and complete. Truncation is inferred by re-tokenising saved text; treating a borderline truncated zero-shot row as complete can only shrink the tested difference, so the test is conservative for T.

5.5 Clarify Training and the Detector

C1–C3 asked whether 300 templated clarification examples can teach the missing behaviour without harming numeric accuracy. Table 5.2 and Fig. 5.7(a) give the automatic answer.

Clarify exact rises from 0% to 29.4–38.2% (family A3a: + 29.4 to + 38.2 percentage points; every p_{Holm} below .001). Invented values fall from 100% to 50.0–66.2% (A3b: − 33.8 to − 50.0 percentage points; every p_{Holm} below .001). C1 and C2 are supported. The examples never delete ET_0 and never show an ambiguous crop or an unknown unit, so the residual invented values include generalisation the training did not demonstrate.

C3 asked whether numeric accuracy falls. On `correct_0p1` over 146 rows, the clarify-trained minus fine-tuned differences are small and none is significant after Holm. The lowest confidence-interval bound is −8.2 percentage points (Phi-3.5, flagged in advance). No loss is detected. Clarify-trained arms exist at seed 42 only, so C1–C3 have no seed-robustness check.

Family A5, the detector, is descriptive. Applied to clarify-trained Llama, it raises safe handling of incomplete questions from 36.8% to 83.8% (risk difference +47.1 [+35.3, +58.8] percentage points; 32 rows caught, 0 reversed). The detector runs only on rows whose gold type is clarify, flags any invented numeric answer on the three scorable clarify categories, and had its missing-ET_0 keywords added after the test rows were inspected. All three properties use information a deployed system would not have. The figure is an oracle-gated upper bound on what these surface rules catch in this test set. The rules are listed in Chapter F.

Table 5.2: Clarify training, per model: seed 42 fine-tuned (FT) against seed 42 clarify-trained (CL), 68 clarify rows, values in per cent and differences in percentage points. Clarify exact (asks and names the missing parameter) is the primary measure; clarify learned (any request that withholds the result) is secondary and untested. Every difference in both tested families (A3a, A3b) has $p_{\text{Holm}} < .001$. Clarify-trained models exist at seed 42 only.

	Clarify exact (primary)			Clarify learned		Invented value		
Model	FT	CL	CL − FT [95% CI]	FT	CL	FT	CL	CL − FT [95% CI]
Llama-3.1-8B	0.0	33.8	+33.8 [+22.1, +45.6]	0.0	36.8	100.0	63.2	−36.8 [−48.5, −25.0]
Mistral-NeMo-12B	0.0	38.2	+38.2 [+26.5, +50.0]	0.0	50.0	100.0	50.0	−50.0 [−61.8, −38.2]
Phi-3.5-mini	0.0	29.4	+29.4 [+19.1, +39.7]	0.0	33.8	100.0	66.2	−33.8 [−45.6, −22.1]
Phi-4-mini	0.0	38.2	+38.2 [+26.5, +50.0]	0.0	41.2	100.0	58.8	−41.2 [−52.9, −29.4]

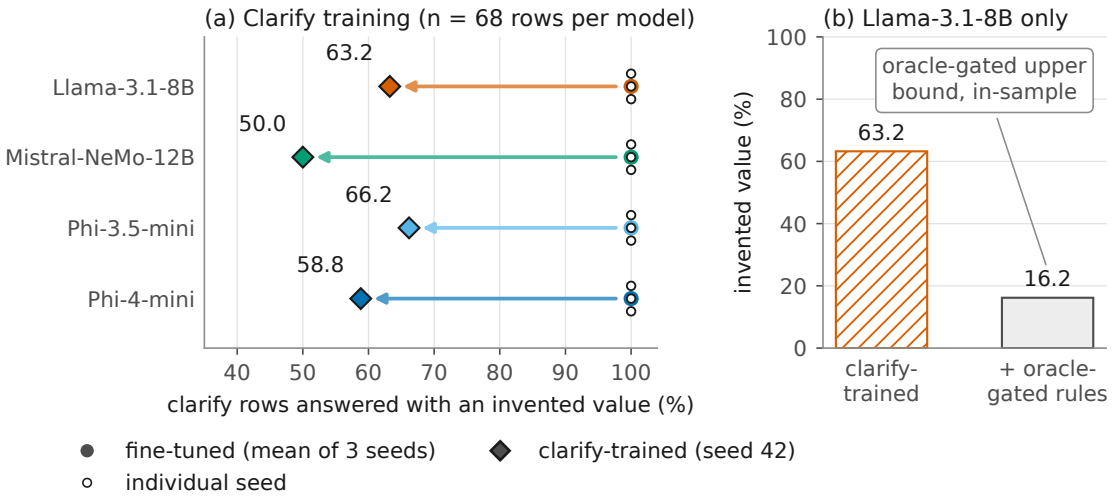


Figure 5.7: (a) Invented-value rate on 68 rows: fine-tuned (circles; three seeds) against clarify-trained (diamonds, seed 42). (b) Llama-3.1-8B only: invented values remaining after clarify training, and after an oracle-gated rule set is applied to those outputs. The second bar is an in-sample upper bound, not a performance estimate; the detector reads gold labels.

5.6 Bengali versus English

H2 said that Bengali trails English after fine-tuning. Table 5.3 and Fig. 5.8 split that claim by question type.

On factual chrF++, English is ahead in every model at seed 42 (mean differences +6.3 to +12.8 points). The registered Wilcoxon test is significant after Holm in 3 of four models (Mistral, Phi-3.5, Phi-4) and robust on all three seeds in 2 of four (the two Phi models). Llama's rank test is not significant ($p_{\text{Holm}} = .526$; rank-biserial $r = 0.23$ [−0.07, 0.52]), while the bootstrap interval of the *mean* difference, +6.3 [+1.1, +12.0] chrF++ points, excludes zero. The paired differences are skewed: a few English rows gain a lot. The registered verdict stands: no significant difference detected for Llama. Mistral's gap is significant at seed 42 only.

On numeric accuracy the picture is different. No unpaired Fisher test on the 73 versus 73 rows is significant at any seed. The confidence intervals of the English-minus-Bengali difference have half-widths of about 15 percentage points; they are too wide to rule out a gap of that size in either direction. There is no consistent sign.

Human language contrasts (family B5, 100 pairs) go the same way: none of eight tests is significant after Holm. Point estimates lean English in the three fine-tuned arms. H2 is therefore only partially supported: a factual English lead that is not universal, a numeric comparison that cannot rule out a moderate gap, and no significant human difference detected.

An exploratory check, added after Part A, classifies the language of the response by script (Fig. 5.9). Zero-shot Mistral-NeMo answered only 30.4% [23.8, 37.0] of Bengali questions in Bengali, and 0.0% of Bengali clarify questions. Every fine-tuned and clarify-trained arm answered at least 98.9% in Bengali. Fine-tuning fixes response language. It also qualifies Mistral's zero-shot confident-error rate: that baseline is measured largely on English answers to Bengali questions.

Table 5.3: Bengali versus English, fine-tuned models, seed 42 (test family A6). Numeric: share of the 146 numeric rows correct within ±0.1 mm, half in each language, compared with an unpaired Fisher exact test. Factual: chrF++ on facts asked in both languages, compared with a Wilcoxon signed-rank test; the difference is in chrF++ points. CIs are 10,000-resample bootstrap intervals. "All seeds" says whether the factual difference is significant after Holm correction at seeds 42, 1337 and 2024.

	Numeric, ±0.1 mm (%), 146 rows				Factual chrF++, 57 paired facts				
Model	bn	en	en − bn [95% CI]	p_{Holm}	bn	en	en − bn [95% CI]	p_{Holm}	all seeds
Llama-3.1-8B	52.1	53.4	+1.4 [−15.1, +17.8]	= 1.000	26.06	32.34	+6.3 [+1.1, +12.0]	= .526	no
Mistral-NeMo-12B	50.7	50.7	0.0 [−16.4, +16.4]	= 1.000	25.63	36.75	+11.1 [+5.1, +17.6]	= .002	no
Phi-3.5-mini	23.3	37.0	+13.7 [−1.4, +28.8]	= .520	18.69	26.05	+7.4 [+2.3, +12.8]	= .001	yes
Phi-4-mini	31.5	35.6	+4.1 [−11.0, +19.2]	= 1.000	18.37	31.22	+12.8 [+6.6, +19.8]	< .001	yes

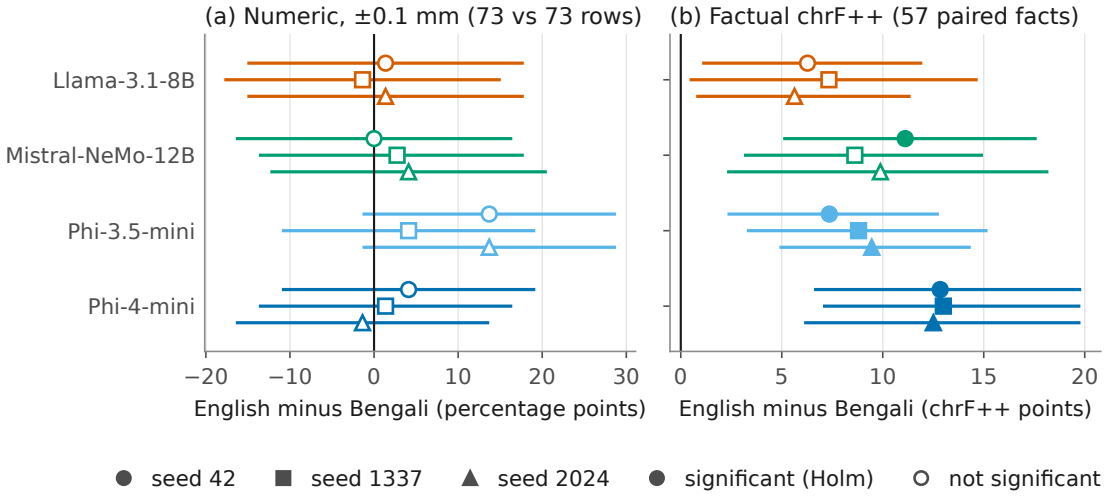


Figure 5.8: English minus Bengali, fine-tuned arms, three seeds. (a) Numeric accuracy within ±0.1 mm, 73 rows per language, unpaired. (b) Factual chrF++, 57 paired facts. Filled markers are significant after Holm; open markers are not. Most numeric intervals include zero. The factual gap favours English and is significant at seed 42 in three of four models.

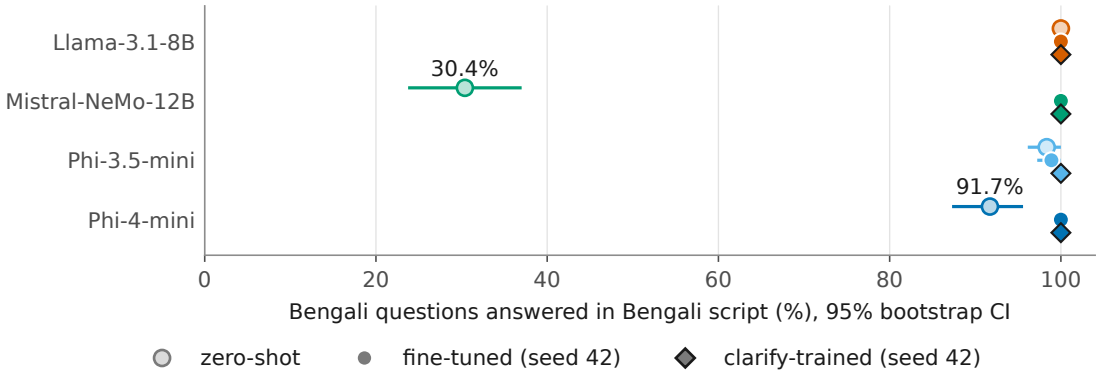


Figure 5.9: Share of Bengali test questions answered in Bengali script (exploratory; 181 questions per arm). Open circles are zero-shot, filled circles fine-tuned (seed 42), diamonds clarify-trained. Mistral zero-shot is the only arm that routinely answers Bengali questions in English; after fine-tuning every arm is at or above 98.9%.

5.7 Human Evaluation

H3 named two primary agreement measures before testing, with a threshold of 0.6: Fleiss' κ on the farmer decision and ordinal Krippendorff's α on the 1–5 rating. Table 5.4 reports both, with bootstrap intervals over the 100 shared outputs. Farmer κ is 0.647 [0.542, 0.740]; ordinal α is 0.721 [0.635, 0.787]. Both clear the threshold. Nominal κ on the raw 1–5 scale is 0.392 [0.321, 0.455]; it treats a 4-versus-5 disagreement as a full disagreement and so understates agreement on an ordinal scale. It is reported for completeness, not as a primary measure. H3 is supported on the pre-specified primary measures.

Leave-one-out ordinal α rises to 0.781 without annotator A1 and falls without any of the others (Fig. 5.10). Sensitivity analyses that drop A1, and that add the 18 authored translations, change no human verdict (Section 4.4.7).

Table 5.5 and Fig. 5.11 are the arm-level human scores on the primary sample. Fine-tuning raises mean rating from 1.96 [1.82, 2.11] to 3.13 [2.91, 3.35] and farmer-yes from 10.4 [6.0, 14.8]% to 42.3 [35.2, 49.5]% (B2-i and B3-i; both p_{Holm} below .001). That corroborates H1. On clarify questions the standard fine-tuned arms are judged unfit to give a farmer on every primary-sample item (0.0% for Llama FT; the Mistral cell is the same 0 of 25). The clarify-trained arm reaches 56.0% [36.0, 76.0] farmer-yes on those questions (B3-ii clarify: + +56.0 [+36.0, +76.0] percentage points, p_{Holm} below .001), which corroborates C1.

The overall clarify-versus-fine-tuned contrast is measure-dependent. Farmer-yes rises by +11.0 [+3.8, +18.1] percentage points and survives Holm (p_{Holm} = .033). Mean rating rises by less and does not (p_{Holm} = .128). The annotator-adjusted model finds the rating contrast significant when it is not corrected for multiplicity; B2's corrected verdict stands. Report the overall human benefit of clarify training as significant on farmer-yes and not on rating after correction.

Human acceptance of numeric answers tracks agreement with the visible reference, not the $\pm$0.1 mm headline. Spearman ρ between absolute error and mean rating is −0.74 [−0.82, −0.65] on 180 fine-tuned numeric outputs (p_{Holm} below .001; 21 multi-crop outputs have no scalar error and are dropped). Fig. 5.12 bins the same outputs. Annotators accepted 97% of exact matches (below 0.005 mm) and 38% of answers 0.005–0.1 mm away; the

Table 5.4: Agreement on the 100 outputs rated by all five annotators, with 95% bootstrap CIs over items. The two primary measures were named before significance testing. Nominal κ treats adjacent ratings (four versus five) as full disagreement and so understates agreement on an ordinal scale; it is reported for completeness. Leave-one-out values were recomputed and reproduce the audit report's all-annotator and without-A1 values.

Measure	Statistic	Value	95% CI
Would give to a farmer	Fleiss' κ (primary)	0.647	[0.542, 0.740]
Would give to a farmer	Krippendorff's α, nominal	0.648	[0.543, 0.741]
Rating 1–5	Krippendorff's α, ordinal (primary)	0.721	[0.635, 0.787]
Rating 1–5	Krippendorff's α, interval	0.746	[0.670, 0.806]
Rating, three bands	Fleiss' κ	0.561	[0.482, 0.633]
Rating 1–5	Fleiss' κ, nominal	0.392	[0.321, 0.455]
Rating, leave one out	ordinal α without A1 / A2 / A3 / A4 / A5	0.781 / 0.685 / 0.729 / 0.716 / 0.687	

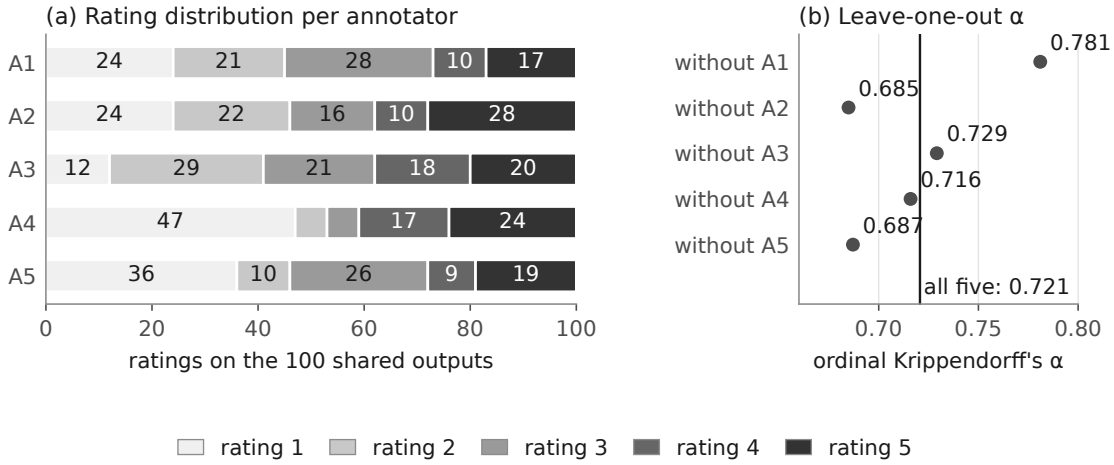


Figure 5.10: (a) Rating distribution of each annotator on the 100 shared outputs. (b) Ordinal α when that annotator is left out; the vertical line is α with all five. Leaving out A1 raises agreement; leaving out any other annotator lowers it.

Table 5.5: Human results per arm on the primary sample of 182 test-set questions, with 95% bootstrap CIs. A shared output's score is the mean of its five ratings and its farmer decision is the majority vote. "Acceptable" means a mean rating of at least four. The clarify column covers the 25 clarify questions in the primary sample. The last column adds the 18 authored Bengali translations (sensitivity sample). Raw means do not adjust for which annotator rated which output; the annotator-adjusted models (B4) leave every contrast's direction and significance unchanged.

Arm	Mean rating (1–5)	Farmer yes (%)	Acceptable (%)	Clarify yes (%)	All 200: yes (%)
Llama zero-shot	1.96 [1.82, 2.11]	10.4 [6.0, 14.8]	7.7 [3.8, 11.5]	0.0	9.5
Llama fine-tuned	3.13 [2.91, 3.35]	42.3 [35.2, 49.5]	40.7 [33.5, 47.8]	0.0	39.0
Llama clarify-trained	3.42 [3.19, 3.63]	53.3 [45.6, 60.4]	50.0 [42.3, 57.1]	56.0	50.0
Mistral fine-tuned	3.05 [2.83, 3.27]	39.6 [32.4, 46.7]	37.9 [30.8, 45.1]	0.0	36.0

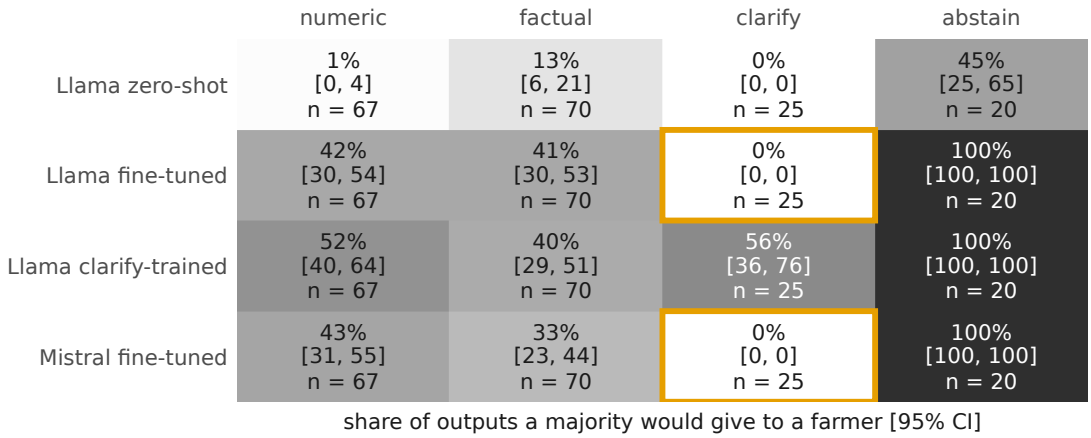


Figure 5.11: Share of outputs a majority would give to a farmer, by arm and question type, primary sample, with 95% CIs. The highlighted cells are incomplete questions: 0% for both standard fine-tuned arms, 56.0% for Llama clarify-trained.

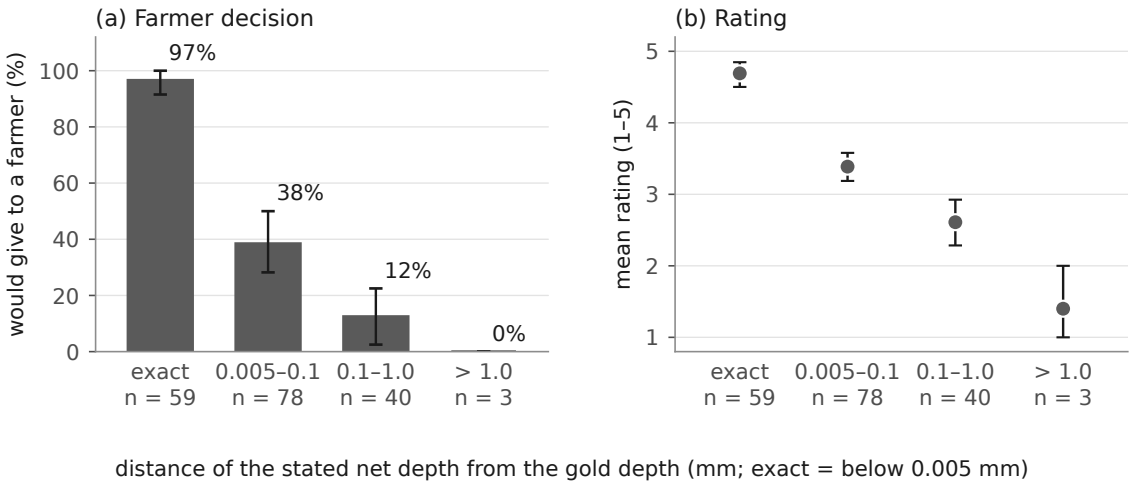


Figure 5.12: Human acceptance (a) and mean rating (b) by distance of the stated net depth from the gold, three fine-tuned arms, primary numeric questions. With the reference visible, raters accepted 97% of exact answers and 38% of answers inside the 0.005–0.1 mm band that the automatic headline still counts as correct.

0.1–1.0 mm bin is accepted 12% of the time. The registered headline, that acceptance tracks ±0.1 mm rather than ±1 mm, holds. The additional finding is that ±0.1 mm is more lenient than the human standard: `exact_0.1` counts as correct many answers raters would not hand over.

Word-level Dice is the automatic factual metric that best tracks the farmer decision (ROC-AUC 0.843 [0.788, 0.892] on 280 factual judgements). Its interval overlaps ROUGE-L and chrF++. Character-level Dice tracks worst. That closes the deferred metric check: chrF++ remains the headline because it was fixed before results, and Dice is the closest of the set to the raters.

5.8 Llama Fine-Tuned versus Mistral Fine-Tuned

The pre-registered comparison of the two large models after fine-tuning found no significant difference. On numeric `correct_0p1`, Mistral minus Llama is −2.1 [−11.0, +6.8] percentage points (McNemar odds ratio 0.88 [0.46, 1.64]), non-significant at all three seeds. Human rating and farmer-yes go the same way (B2-iii, B3-iii), and the annotator-adjusted models leave direction and significance unchanged. The four fine-tuned models do differ as a set (Friedman $W = 0.09$ [0.05, 0.15]); the pre-specified tier contrast, {Llama, Mistral} minus {Phi-3.5, Phi-4}, is significant and robust. Fig. 5.13 places these effects next to the others.

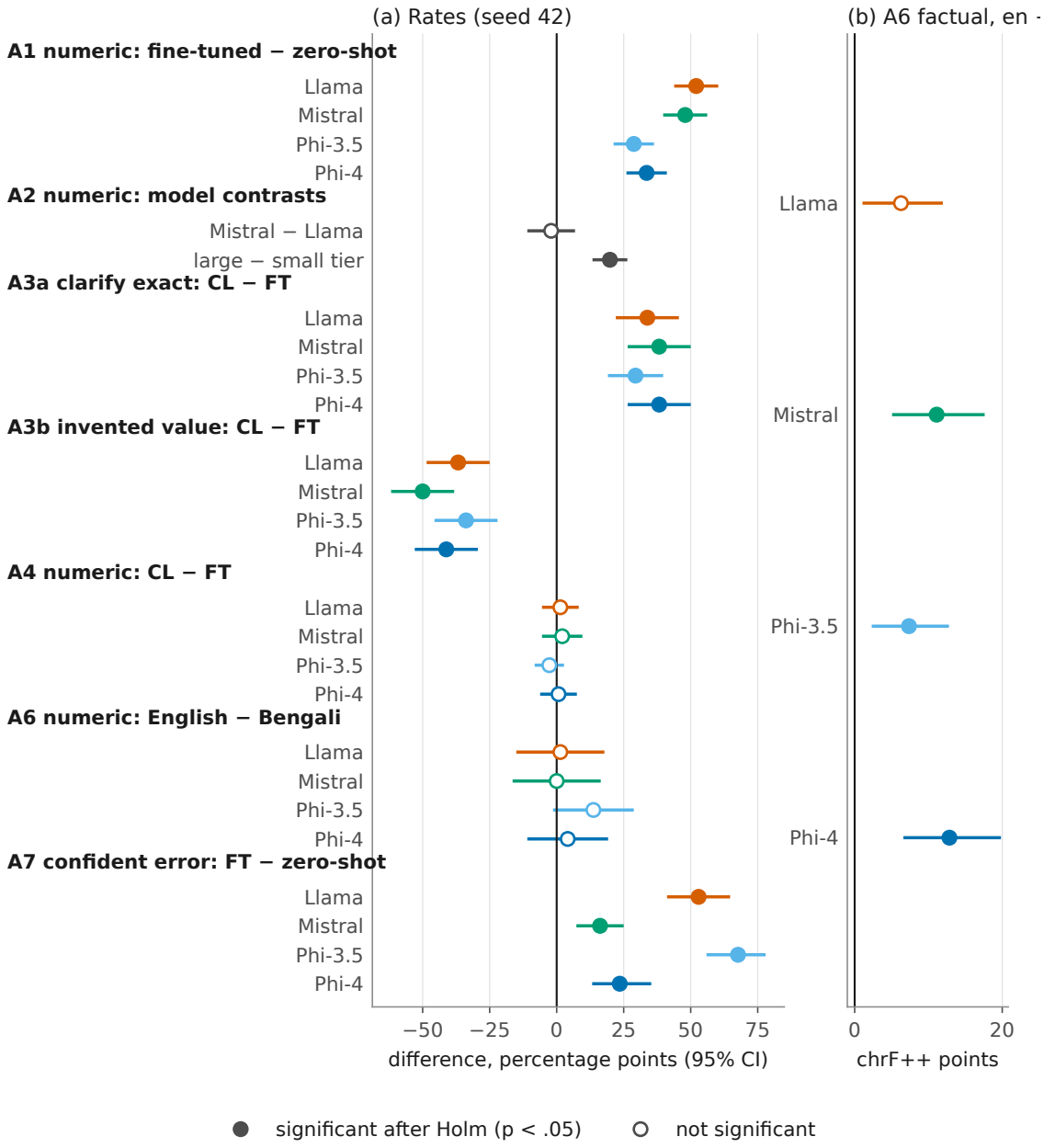


Figure 5.13: Primary automatic effects at seed 42 with 95% CIs. (a) Rate differences in percentage points. (b) Factual English-minus-Bengali chrF++. Filled markers are significant after Holm. A1, A3, A7 and the tier contrast sit away from zero; A2 (Mistral minus Llama), A4 (clarify-trained minus fine-tuned numeric) and A6 numeric include zero.

5.9 Hypothesis Summary

Table 5.6 is the register of verdicts. Nothing in it reversed a pre-specified direction. Three Stage 6 readings were qualified once tested: the factual English lead is not universal; $\pm$0.1 mm is more lenient than human acceptance; and the detector figure is not a performance estimate. Clarify training's overall human benefit is significant on farmer-yes and not on rating after Holm. Mistral zero-shot answered most Bengali questions in English, which the Stage 6 metrics did not distinguish from Bengali answers.

Table 5.6: Hypothesis verdicts under the analysis plan fixed before significance testing (plan v3). Automatic tests use seed 42 with Holm correction within each family; human tests use the primary sample of 182 test-set questions. pp = percentage points; intervals are 95% bootstrap CIs.

ID	Claim	Tests	Verdict	Key numbers
H1	Fine-tuning beats zero-shot	A1, B2-i, B3-i	Supported; robust on all three seeds	A1: 28.8 to 52.1 pp, every p_{Holm} < .001; farmer yes +31.9 [+24.7, +39.0] pp
H2	Bengali trails English	A6, B5	Partially supported	factual: significant in 3 of four models at seed 42, 2 of four on all seeds; numeric: 0 of 12 tests; human: 0 of 8
H3	Annotators agree	B6	Supported on the primary measures	farmer κ 0.647 [0.542, 0.740]; ordinal α 0.721 [0.635, 0.787]
T	Fine-tuning turns invented values into confident errors	A7	Supported in all four models on all seeds; size depends on the base model	16.2 to 67.6 pp; fine-tuned strict rate 98.5–100.0%
C1	Clarify training teaches clarification	A3a, B3-ii	Supported	+29.4 to +38.2 pp; human clarify cell +56.0 [+36.0, +76.0] pp
C2	Clarify training reduces invented values	A3b	Supported	−33.8 to −50.0 pp
C3	Clarify training does not harm numeric accuracy	A4	No loss detected	0 of four significant; lowest CI bound −8.2 pp
D	A detector adds safety	A5	Descriptive only: oracle-gated upper bound	+47.1 [+35.3, +58.8] pp, no test
M	Llama FT differs from Mistral FT	A2, B2-iii, B3-iii, B4	No significant difference detected	−2.1 [−11.0, +6.8] pp; farmer yes −2.7 [−9.3, +3.8] pp

5.10 Threats to Validity

Construct. The confident-error definition depends on a response classifier, a parser and a re-tokenised length. Decode-then-encode is not perfectly idempotent; the threshold was kept as registered and the bias is conservative for T. The Bengali hedge list under-detects zero-shot hedging and does not affect any fine-tuned row we checked. Numeric "accuracy" is agreement with this system's calculator, including its unclamped zero depths and its project-specific rainfall rule. Human ratings were made with the gold visible and with a rubric that tells raters to score invented values 1, so they corroborate the automatic measures rather than independently testing field usefulness. Annotators are not agricultural experts.

Internal. The twelve fine-tuned runs share data and base weights; seeds measure training randomness, not independent samples. Clarify-trained arms exist at one seed, so C1–C3 cannot be called robust. Three metric definitions were changed after results were seen; each was checked against a gold-oracle arm whose score after the changes is 1.000, and whose score before them is not in any surviving file. The analysis plan was fixed before significance tests and after the descriptive results; we do not call it pre-registered. A5 reads gold labels. The semantic leakage review is still pending (**PENDING REVIEW:** semantic leakage review, 60 pairs). Digitisation of the calendars has not been independently checked (**STAGE 8B, NOT YET RUN:** digitisation check).

External. All questions are synthetic. No farmer took part. Four open-weight models, none larger than 12B, and no commercial model. Abstention is measured on easy seed out-of-domain rows; 60 adversarial rows are unscored. There is no system prompt and no calculator in the loop, so the study measures what fine-tuning does on its

own, not what an instructed or tool-using system would do. The prompted baseline is **STAGE 8B, NOT YET RUN:** prompted baseline. Retrieval was not compared. Results are for this coefficient table, this rainfall rule and this template style.

Statistical. Holm correction is within family, not across the whole plan. Numeric language tests are unpaired and underpowered; their intervals do not rule out gaps of about 15 percentage points. MixedLM treats the rating as interval; the Bayesian farmer-yes model yields credible intervals, not *p*-values. We never read a non-significant test as equivalence.

Chapter 6

Conclusion

This chapter answers the research questions, restates what the study can and cannot claim, and lists the work that should come next.

6.1 Answers to the Research Questions

RQ1 / H1. Fine-tuning on complete-input examples teaches the models this system's conventions and its arithmetic. No base model states a correct crop coefficient or applies the tiered effective-rainfall rule. After fine-tuning, K_c F1 is 0.542–0.761 and effective-rainfall F1 is 0.930–0.999. Accuracy within $\pm$0.1 mm on the 116 single-crop rows rises from 0.0–3.4% to 38.8–69.3%, a gain that is significant after Holm in every model at every seed. Human ratings move with the automatic scores. H1 is supported. The gain is agreement with this calculator and this template, not a claim about reasoning.

RQ2 / T. When a required input is missing, fine-tuned models invent it. They do not ask and they do not refuse. Abstention, which was 3.6% of the training corpus, is learned completely (100% on the 34 seed out-of-domain rows in all twelve runs). Clarification, which was 0.0% of training, is learned not at all (0% named clarification in all twelve runs). Base models also invent values, but often by running to the token limit (13.2–66.2% of clarify responses) or by saying that a value was assumed (15–42% of invented answers). After fine-tuning, 98.5–100.0% of clarify responses are complete, unhedged, confident errors. The paired test is significant in every model at every seed. T is supported. The size of the change depends on the base model. Annotators would not hand any of the evaluated standard fine-tuned models' answers to incomplete questions to a farmer.

RQ3 / C1–C3. 300 templated clarification examples raise named clarification to 29.4–38.2% and cut invented values from 100% to 50.0–66.2%, with no significant numeric loss detected (lowest CI bound −8.2 percentage points). Human farmer-yes on clarify questions rises by +56.0 [+36.0, +76.0] percentage points. C1 and C2 are supported; C3 is reported as no loss detected. The overall human rating gain does not survive Holm. Clarify-trained arms exist at one seed. A post-hoc detector that reads gold labels can push Llama's safe

handling from 36.8% to 83.8%; that figure is an oracle-gated upper bound, not a mitigation result.

RQ4 / H2. Bengali trails English on factual text in every model (6 to 13 chrF++ points), significantly after Holm in three of four at seed 42 and in two of four on all seeds. Llama's registered rank test is not significant. Numeric English–Bengali differences are not significant in any model or seed; the intervals are too wide to rule out a gap of about 15 percentage points. Human language contrasts are not significant after Holm. H2 is partially supported. Fine-tuning does fix response language: zero-shot Mistral answered only 30.4% of Bengali questions in Bengali; every fine-tuned arm answered at least 98.9%.

RQ5 / H3. On the two measures named before testing, agreement clears the threshold of 0.6: farmer κ is 0.647 [0.542, 0.740] and ordinal α is 0.721 [0.635, 0.787]. H3 is supported on those measures. Nominal κ on the raw 1–5 scale is lower, as expected for an ordinal instrument scored nominally. Human acceptance of numeric answers tracks exact agreement with the visible reference (97% of exact matches, 38% of the 0.005–0.1 mm band), so the automatic $\pm$0.1 mm headline is more lenient than the raters.

Llama fine-tuned and Mistral fine-tuned do not differ significantly on the pre-specified tests. The four fine-tuned models do differ as a set; the large-versus-small tier contrast is significant and robust.

6.2 What This Thesis Does Not Claim

It does not claim that large language models cannot ask for missing information. It claims that standard LoRA fine-tuning on this corpus, with no clarification examples and no instruction to ask, produces confident invented answers. Targeted examples change that in part. A prompt that tells the base model to ask was not run (**STAGE 8B, NOT YET RUN:** prompted baseline).

It does not claim that the models have learned irrigation. They have learned this coefficient table, this rainfall rule and this answer template. Most training numeric rows come from the same generator as the test rows.

It does not claim that the testbed is ready for farmers. No farmer took part. 60 test rows are unscored. The raters are the authors, not extension officers. We do not recommend deploying any model studied here.

It does not claim that a non-significant language or model comparison is equivalence. The intervals are reported so that a reader can see what is ruled out.

6.3 Future Work

The highest-leverage experiment that is already designed and not yet run is the prompted baseline: the same four models, with and without fine-tuning, given a system prompt that states the formula, the coefficient table and an instruction to ask for missing values

(STAGE 8B, NOT YET RUN: prompted baseline). That experiment separates what weights teach from what an instruction can achieve.

Four other items belong to the same supplementary round: class-level scoring of the 60 excluded rows; a two-person digitisation check of the calendars against their images; a check of the coefficient table against FAO-56 and the Bangladesh rice study; and the two external raters on the shared outputs.

A detector that reads only the question, not the gold type or category, is future work if anyone wants a runtime guard. So is putting a calculator in the loop: a model that invents a missing input will pass the invented value downstream, and measuring that failure with a tool present would show whether the behaviour survives the obvious engineering fix.

On the data side, paraphrase and back-translation were never run; a farmer-authored test set does not exist; and the semantic leakage sample is still unlabelled (PENDING REVIEW: semantic leakage review). Any of those would tighten the external-validity claim.

Clarify training should be repeated at more than one seed, and should include the missing-ET_0, ambiguous-crop and unknown-unit cases that the first 300 examples omitted. Whether a small number of such examples is enough, or whether the model only copies the template, is an open measurement.

6.4 Concluding Remarks

Fine-tuning is the usual way to make a general model useful in a narrow domain, and accuracy is the usual way to decide that it worked. On this irrigation testbed both statements are true and incomplete. The models learn the conventions and the arithmetic. They also learn to finish an answer when the question does not contain one. The unfinished, hedged errors of the base models become short, well-formed volumes. That change is the finding. A few hundred examples that show the other behaviour undo part of it, and nothing we measured suggests that the arithmetic has to be sacrificed to do so. Until asking for a missing input is taught, tested and required, a fine-tuned agricultural assistant of this kind is not safe to hand to a farmer.

Bibliography

[1] Bangladesh Agricultural Research Council. "Crop calendar." Online application; page title "Crop Calender Produced by BARC", Accessed: Sep. 24, 2026. [Online]. Available: http://apps.barc.gov.bd/cropcalendar/

[2] R. G. Allen, L. S. Pereira, D. Raes, and M. Smith, *Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements* (FAO Irrigation and Drainage Paper 56). Rome: Food and Agriculture Organization of the United Nations, 1998, ISBN: 92-5-104219-5. Accessed: Sep. 24, 2026. [Online]. Available: https://www.fao.org/4/x0490e/x0490e00.htm

[3] M. Shamsudduha, R. G. Taylor, M. I. Haq, S. Nowreen, A. Zahid, and K. M. U. Ahmed, "The Bengal Water Machine: Quantified freshwater capture in Bangladesh," *Science*, vol. 377, no. 6612, pp. 1315–1319, 2022. DOI: 10.1126/science.abm4730

[4] N. Singh et al., *Farmer.Chat: Scaling AI-Powered Agricultural Services for Smallholder Farmers*, arXiv preprint, 2024. arXiv: 2409.08916.

[5] M. R. Ameen, A. Islam, F. Aktar, and M. S. Rafat, *KrishokBondhu: A Retrieval-Augmented Voice-Based Agricultural Advisory Call Center for Bengali Farmers*, arXiv preprint, 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) (arXiv journal reference), 2025. DOI: 10.1109/QPAIN69676.2026.11546653 arXiv: 2510.18355.

[6] A. Balaguer et al., *RAG vs Fine-tuning: Pipelines, Tradeoffs, and a Case Study on Agriculture*, arXiv preprint, 2024. arXiv: 2401.08406.

[7] D. J. Samuel, I. Skarga-Bandurova, D. Sikolia, and M. Awais, *AgroLLM: Connecting Farmers and Agricultural Practices through Large Language Models for Enhanced Knowledge Transfer and Practical Application*, arXiv preprint, 2025. arXiv: 2503.04788.

[8] A. Tzachor et al., "Large language models and agricultural extension services," *Nature Food*, vol. 4, no. 11, pp. 941–948, 2023. DOI: 10.1038/s43016-023-00867-x

[9] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," in *International Conference on Learning Representations (ICLR)*, 2022. arXiv: 2106.09685.

[10] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv: 2305.14314.

[11] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang, *Why Language Models Hallucinate*, arXiv preprint, 2025. arXiv: 2509.04664.

[12] Z. Gekhman et al., "Does fine-tuning LLMs on new knowledge encourage hallucinations?" In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Miami, Florida, USA: Association for Computational Linguistics, 2024, pp. 7765–7784. DOI: 10.18653/v1/2024.emnlp-main.444 [Online]. Available: https://aclanthology.org/2024.emnlp-main.444/

[13] B. Z. Li, B. Kim, and Z. Wang, *QuestBench: Can LLMs ask the right question to acquire information in reasoning tasks?* arXiv preprint, NeurIPS 2025 Datasets and Benchmarks Track (arXiv journal reference), 2025. arXiv: 2503.22674.

[14] P. Kirichenko, M. Ibrahim, K. Chaudhuri, and S. J. Bell, *AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions*, arXiv preprint, 2025. arXiv: 2506.09038.

[15] N. G. Dastane, *Effective Rainfall in Irrigated Agriculture* (FAO Irrigation and Drainage Paper 25). Rome: Food and Agriculture Organization of the United Nations, 1978, Web edition, copyright FAO 1978, ISBN: 92-5-100272-X. Accessed: Sep. 24, 2026. [Online]. Available: https://www.fao.org/4/x5560e/x5560e00.htm

[16] A. Vaswani et al., "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. arXiv: 1706.03762.

[17] J. Wei et al., "Finetuned Language Models Are Zero-Shot Learners," in *International Conference on Learning Representations (ICLR)*, 2022. arXiv: 2109.01652.

[18] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv: 2203.02155.

[19] A. Grattafiori et al., *The Llama 3 Herd of Models*, arXiv preprint, 2024. arXiv: 2407.21783.

[20] Mistral AI. "Mistral NeMo," Accessed: Sep. 24, 2026. [Online]. Available: https://mistral.ai/news/mistral-nemo

[21] M. Abdin et al., *Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone*, arXiv preprint, 2024. arXiv: 2404.14219.

[22] Microsoft et al., *Phi-4-Mini Technical Report: Compact yet Powerful Multimodal Language Models via Mixture-of-LoRAs*, arXiv preprint, 2025. arXiv: 2503.01743.

[23] N. Houlsby et al., "Parameter-Efficient Transfer Learning for NLP," in *Proceedings of the International Conference on Machine Learning (ICML), PMLR 97*, 2019. arXiv: 1902.00751.

[24] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv: 2005.11401.

[25] M. Popović, "ChrF: Character n-gram F-score for automatic MT evaluation," in *Proceedings of the Tenth Workshop on Statistical Machine Translation*, Lisbon, Portugal: Association for Computational Linguistics, 2015, pp. 392–395. DOI: 10.18653/v1/W15-3049 [Online]. Available: https://aclanthology.org/W15-3049/

[26] M. Popović, "ChrF++: Words helping character n-grams," in *Proceedings of the Second Conference on Machine Translation*, Copenhagen, Denmark: Association for Computational Linguistics, 2017, pp. 612–618. DOI: 10.18653/v1/W17-4770 [Online]. Available: https://aclanthology.org/W17-4770/

[27] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Text Summarization Branches Out*, Barcelona, Spain: Association for Computational Linguistics, 2004, pp. 74–81. [Online]. Available: https://aclanthology.org/W04-1013/

[28] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in *International Conference on Learning Representations (ICLR)*, 2020. arXiv: 1904.09675.

[29] J. Novikova, O. Dušek, A. Cercas Curry, and V. Rieser, "Why we need new evaluation metrics for NLG," in *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, Copenhagen, Denmark: Association for Computational Linguistics, 2017, pp. 2241–2252. DOI: 10.18653/v1/D17-1238 [Online]. Available: https://aclanthology.org/D17-1238/

[30] J. L. Fleiss, "Measuring nominal scale agreement among many raters.," *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971. DOI: 10.1037/h0031619

[31] A. F. Hayes and K. Krippendorff, "Answering the Call for a Standard Reliability Measure for Coding Data," *Communication Methods and Measures*, vol. 1, no. 1, pp. 77–89, 2007. DOI: 10.1080/19312450709336664

[32] J. R. Landis and G. G. Koch, "The Measurement of Observer Agreement for Categorical Data," *Biometrics*, vol. 33, no. 1, p. 159, 1977. DOI: 10.2307/2529310

[33] Q. McNemar, "Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. DOI: 10.1007/BF02295996

[34] F. Wilcoxon, "Individual Comparisons by Ranking Methods," *Biometrics Bulletin*, vol. 1, no. 6, p. 80, 1945. DOI: 10.2307/3001968

[35] M. Friedman, "The Use of Ranks to Avoid the Assumption of Normality Implicit in the Analysis of Variance," *Journal of the American Statistical Association*, vol. 32, no. 200, pp. 675–701, 1937. DOI: 10.1080/01621459.1937.10503522

[36] W. G. COCHRAN, "THE COMPARISON OF PERCENTAGES IN MATCHED SAMPLES," *Biometrika*, vol. 37, no. 3-4, pp. 256–266, 1950. DOI: 10.1093/biomet/37.3-4.256

[37] B. Efron, "Bootstrap Methods: Another Look at the Jackknife," *The Annals of Statistics*, vol. 7, no. 1, 1979. DOI: 10.1214/aos/1176344552

[38] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[39] R. Dror, G. Baumer, S. Shlomov, and R. Reichart, "The hitchhiker's guide to testing statistical significance in natural language processing," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Melbourne, Australia: Association for Computational Linguistics, 2018, pp. 1383–1392. DOI: 10.18653/v1/P18-1128 [Online]. Available: https://aclanthology.org/P18-1128/

[40] B. Silva, L. Nunes, R. Estevão, V. Aski, and R. Chandra, *GPT-4 as an Agronomist Assistant? Answering Agriculture Exams Using Large Language Models*, arXiv preprint, 2023. arXiv: 2310.06225.

[41] J. Kpodo, P. Kordjamshidi, and A. P. Nejadhashemi, "AgXQA: A benchmark for advanced Agricultural Extension question answering," *Computers and Electronics in Agriculture*, vol. 225, p. 109 349, 2024. DOI: 10.1016/j.compag.2024.109349

[42] M. A. Hossain, N. Subhan, M. R. Mahi, and J. F. Nabila, *Cost-Efficient Cross-Lingual Retrieval-Augmented Generation for Low-Resource Languages: A Case Study in Bengali Agricultural Advisory*, arXiv preprint, 2026. arXiv: 2601.02065.

[43] K. R. I. Reza, S. T. Nimi, and O. I. Shahid, *KrishokChat: A Provenance-Traceable Multi-Task Bengali Agricultural Benchmark with Safety-Critical Chemical Advisory*, arXiv preprint, 2026. arXiv: 2606.29243.

[44] Z. Ji et al., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. DOI: 10.1145/3571730

[45] L. Huang et al., "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, 2025. DOI: 10.1145/3703155

[46] K. Kang, E. Wallace, C. Tomlin, A. Kumar, and S. Levine, "Unfamiliar finetuning examples control how language models hallucinate," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 3600–3612, ISBN: 979-8-89176-189-6. DOI: 10.18653/v1/2025.naacl-long.183 [Online]. Available: https://aclanthology.org/2025.naacl-long.183/

[47] H. Zhang et al., "R-tuning: Instructing large language models to say 'I don't know'," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Mexico City, Mexico: Association for Computational Linguistics, 2024, pp. 7113–7139. DOI: 10.18653/v1/2024.naacl-long.394 [Online]. Available: https://aclanthology.org/2024.naacl-long.394/

[48] S. Kadavath et al., *Language Models (Mostly) Know What They Know*, arXiv preprint, 2022. arXiv: 2207.05221.

[49] P. Rajpurkar, R. Jia, and P. Liang, "Know what you don't know: Unanswerable questions for SQuAD," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Melbourne, Australia: Association for Computational Linguistics, 2018, pp. 784–789. DOI: 10.18653/v1/P18-2124 [Online]. Available: https://aclanthology.org/P18-2124/

[50] A. Kamath, R. Jia, and P. Liang, "Selective question answering under domain shift," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online: Association for Computational Linguistics, 2020, pp. 5684–5696. DOI: 10 . 18653 / v1 / 2020 . acl - main . 503 [Online]. Available: https://aclanthology.org/2020.acl-main.503/

[51] B. Wen et al., *Know Your Limits: A Survey of Abstention in Large Language Models*, arXiv preprint, 2024. arXiv: 2407.18418.

[52] F. Brahman et al., "The Art of Saying No: Contextual Noncompliance in Language Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv: 2407.12043.

[53] C. Fan, M. Li, L. Sun, and T. Zhou, *Missing Premise exacerbates Overthinking: Are Reasoning Models losing Critical Thinking Skill?* arXiv preprint, 2025. arXiv: 2504.06514.

[54] S. Rao and H. Daumé III, "Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Melbourne, Australia: Association for Computational Linguistics, 2018, pp. 2737–2746. DOI: 10.18653/v1/P18-1255 [Online]. Available: https://aclanthology.org/P18-1255/

[55] M. Aliannejadi, H. Zamani, F. Crestani, and W. B. Croft, "Asking Clarifying Questions in Open-Domain Information-Seeking Conversations," in *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2019, pp. 475–484. DOI: 10.1145/3331184.3331265 arXiv: 1907.06554.

[56] S. Min, J. Michael, H. Hajishirzi, and L. Zettlemoyer, "AmbigQA: Answering ambiguous open-domain questions," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online: Association for Computational Linguistics, 2020, pp. 5783–5797. DOI: 10.18653/v1/2020.emnlp-main.466 [Online]. Available: https://aclanthology.org/2020.emnlp-main.466/

[57] L. Kuhn, Y. Gal, and S. Farquhar, *CLAM: Selective Clarification for Ambiguous Questions with Generative Language Models*, arXiv preprint, 2022. arXiv: 2212.07769.

[58] M. J. Zhang and E. Choi, "Clarify when necessary: Resolving ambiguity through interaction with LMs," in *Findings of the Association for Computational Linguistics: NAACL 2025*, Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 5541–5558, ISBN: 979-8-89176-195-7. DOI: 10.18653/v1/2025.findings-naacl.306 [Online]. Available: https://aclanthology.org/2025.findings-naacl.306/

[59] M. J. Q. Zhang, W. B. Knox, and E. Choi, *Modeling Future Conversation Turns to Teach LLMs to Ask Clarifying Questions*, arXiv preprint, 2024. arXiv: 2410.13788.

[60] C. Andukuri, J.-P. Fränken, T. Gerstenberg, and N. D. Goodman, *STaR-GATE: Teaching Language Models to Ask Clarifying Questions*, arXiv preprint, 2024. arXiv: 2403.19154.

[61] J. J. Wu, M. Chaudhary, D. Abrahamyan, A. Khaku, A. Wei, and F. H. Fard, *Can Code Language Models Learn Clarification-Seeking Behaviors?* arXiv preprint, 2025. arXiv: 2504.16331.

[62] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, "The state and fate of linguistic diversity and inclusion in the NLP world," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online: Association for Computational Linguistics, 2020, pp. 6282–6293. DOI: 10.18653/v1/2020.acl-main.560 [Online]. Available: https://aclanthology.org/2020.acl-main.560/

[63] A. Bhattacharjee et al., "BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla," in *Findings of the Association for Computational Linguistics: NAACL 2022*, Seattle, United States: Association for Computational Linguistics, 2022, pp. 1318–1327. DOI: 10.18653/v1/2022.findings-naacl.98 [Online]. Available: https://aclanthology.org/2022.findings-naacl.98/

[64] M. Kabir, M. S. Islam, M. T. R. Laskar, M. T. Nayeem, M. S. Bari, and E. Hoque, "BenLLM-eval: A comprehensive evaluation into the potentials and pitfalls of large language models on Bengali NLP," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italia: ELRA and ICCL, 2024, pp. 2238–2252. [Online]. Available: https://aclanthology.org/2024.lrec-main.201/

[65] S. Bhowmik, T. T. Dipto, M. S. Islam, S. Hsu, and T. Reasat, *Script Fragmentation and Format: What Drives the English-Bengali Performance Gap in Open LLMs?* arXiv preprint, 2025. arXiv: 2507.23248.

[66] A. Petrov, E. L. Malfa, P. H. S. Torr, and A. Bibi, "Language Model Tokenizers Introduce Unfairness Between Languages," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv: 2305.15425.

[67] O. Ahia et al., "Do all languages cost the same? tokenization in the era of commercial language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Singapore: Association for Computational Linguistics, 2023, pp. 9904–9923. DOI: 10.18653/v1/2023.emnlp-main.614 [Online]. Available: https://aclanthology.org/2023.emnlp-main.614/

[68] P. Lewis, P. Stenetorp, and S. Riedel, "Question and answer test-train overlap in open-domain question answering datasets," in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, Online: Association for Computational Linguistics, 2021, pp. 1000–1008. DOI: 10.18653/v1/2021.eacl-main.86 [Online]. Available: https://aclanthology.org/2021.eacl-main.86/

[69] I. Magar and R. Schwartz, "Data contamination: From memorization to exploitation," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Dublin, Ireland: Association for Computational Linguistics, 2022, pp. 157–165. DOI: 10.18653/v1/2022.acl-short.18 [Online]. Available: https://aclanthology.org/2022.acl-short.18/

[70] O. Sainz, J. Campos, I. García-Ferrero, J. Etxaniz, O. L. de Lacalle, and E. Agirre, "NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore: Association for Computational Linguistics, 2023, pp. 10 776–10 787. DOI: 10.18653/v1/2023.findings-emnlp.722 [Online]. Available: https://aclanthology.org/2023.findings-emnlp.722/

[71] S. Golchin and M. Surdeanu, "Time Travel in LLMs: Tracing Data Contamination in Large Language Models," in *International Conference on Learning Representations (ICLR)*, 2024. arXiv: 2308.08493.

[72] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100 804, 2023. DOI: 10.1016/j.patter.2023.100804 arXiv: 2207.07048.

[73] C. van der Lee, A. Gatt, E. van Miltenburg, and E. Krahmer, "Human evaluation of automatically generated text: Current trends and best practice guidelines," *Computer Speech & Language*, vol. 67, p. 101 151, 2021. DOI: 10.1016/j.csl.2020.101151

[74] Unsloth AI. "Unsloth." GitHub repository, Accessed: Sep. 24, 2026. [Online]. Available: https://github.com/unslothai/unsloth

[75] Hugging Face. "TRL: Transformer reinforcement learning." GitHub repository, Accessed: Sep. 24, 2026. [Online]. Available: https://github.com/huggingface/trl

[76] T. Wolf et al., "Transformers: State-of-the-art natural language processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Online: Association for Computational Linguistics, 2020, pp. 38–45. DOI: 10.18653/v1/2020.emnlp-demos.6 [Online]. Available: https://aclanthology.org/2020.emnlp-demos.6/

[77] N. Reimers and I. Gurevych, "Making monolingual sentence embeddings multilingual using knowledge distillation," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online: Association for Computational Linguistics, 2020, pp. 4512–4525. DOI: 10.18653/v1/2020.emnlp-main.365 [Online]. Available: https://aclanthology.org/2020.emnlp-main.365/

[78] J. Johnson, M. Douze, and H. Jegou, "Billion-Scale Similarity Search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021. DOI: 10.1109/TBDATA.2019.2921572

[79] M. Post, "A call for clarity in reporting BLEU scores," in *Proceedings of the Third Conference on Machine Translation: Research Papers*, Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 186–191. DOI: 10.18653/v1/W18-6319 [Online]. Available: https://aclanthology.org/W18-6319/

[80] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423 [Online]. Available: https://aclanthology.org/N19-1423/

Appendix A

Worked Examples

Fig. 5.5 in Chapter 5 reprints one incomplete question in both languages, with the three Phi-3.5-mini arms side by side. The model text is verbatim from `reports/s6/predictions/phi35_mini_`. This appendix records why those two rows were chosen and where the rest of the generations live.

Both rows are locked test items whose gold type is clarify because the field unit is not in the conversion table (English "dron"; Bengali গাঙা). The correct response is to ask for the field size. They are not a random sample: they were picked so that a reader can see, on the same question, the three forms discussed in the body—a truncated zero-shot answer, a complete invented volume, and a clarify-trained request.

Every scored generation used in this thesis is stored under `reports/s6/predictions/`, one JSONL file per arm, keyed by `row_id`. The human-study answers to the 18 authored Bengali translations are under `reports/s7/predictions_translated_bn/`. Regenerating a figure or a metric does not require calling a model.

Appendix B

Per-Seed Results

Table 5.1 reports fine-tuned scores as the mean of three seeds. Table B.1 gives the underlying single-run values for every scored arm, including the four clarify-trained models that exist only at seed 42. Family A1 and family A7 remain significant after Holm correction at all three seeds in every model; family A6 factual is robust only for Phi-3.5-mini and Phi-4-mini (Chapter 5).

Table B.1: Per-arm headline scores on the locked test set (one evaluation run each). Numeric: share of 116 single-crop rows within ± 0.1 mm. Clarify exact and invented value: 68 rows. Abstain: 34 seed out-of-domain rows. chrF++: 114 factual rows. Source: `reports/s6/summary_by_arm.csv`.

Model	Arm	± 0.1 mm (%)	Clarify exact (%)	Invented (%)	Abstain (%)	chrF++
Llama-3.1-8B	zero-shot	0.9	0.0	100.0	14.7	11.9
Llama-3.1-8B	FT 42	66.4	0.0	100.0	100.0	29.2
Llama-3.1-8B	FT 1337	69.8	0.0	100.0	100.0	29.5
Llama-3.1-8B	FT 2024	71.6	0.0	100.0	100.0	29.8
Llama-3.1-8B	CL 42	68.1	33.8	63.2	100.0	28.9
Mistral-NeMo	zero-shot	3.4	1.5	97.1	14.7	8.1
Mistral-NeMo	FT 42	63.8	0.0	100.0	100.0	31.2
Mistral-NeMo	FT 1337	70.7	0.0	100.0	100.0	29.3
Mistral-NeMo	FT 2024	66.4	0.0	100.0	100.0	30.3
Mistral-NeMo	CL 42	66.4	38.2	50.0	100.0	30.7
Phi-3.5-mini	zero-shot	1.7	2.9	98.5	20.6	8.8
Phi-3.5-mini	FT 42	37.9	0.0	100.0	100.0	22.4
Phi-3.5-mini	FT 1337	40.5	0.0	100.0	100.0	21.6
Phi-3.5-mini	FT 2024	37.9	0.0	100.0	100.0	20.8
Phi-3.5-mini	CL 42	34.5	29.4	66.2	100.0	21.4
Phi-4-mini	zero-shot	0.0	2.9	98.5	8.8	8.6
Phi-4-mini	FT 42	42.2	0.0	100.0	100.0	24.8
Phi-4-mini	FT 1337	44.0	0.0	100.0	100.0	23.9
Phi-4-mini	FT 2024	42.2	0.0	100.0	100.0	23.1
Phi-4-mini	CL 42	43.1	38.2	58.8	100.0	23.9

Appendix C

Per-Category Results

The hardened test set mixes seed rows with six adversarial categories (Table 4.4). Table C.1 splits the headline scores of the human-study arms, plus Llama zero-shot, by those categories. Missing-parameter, misspelled-crop and unit-ambiguity rows that are labelled clarify are the probe for invented values; seed out-of-domain rows are the probe for abstention. Multi-crop numeric rows are scored only when every gold depth is within 0.1 mm; no arm in this table meets that bar.

Table C.1: Per-category scores for the four human-study arms (and Llama zero-shot). Numeric exact-0.1 is the share of that category's numeric rows within ± 0.1 mm; blank cells are not defined for that category. Source: `reports/s6/per_category.csv`. The 40 adversarial out-of-domain and 20 conflicting-value rows are excluded from scoring and do not appear.

Arm	Category	n	± 0.1 mm (%)	Clarify exact (%)	Invented (%)	Abstain (%)
Llama ZS	seed	182	2.9			14.7
Llama ZS	code switched	40	0.0			
Llama ZS	misspelled crop	40	0.0	0.0	100.0	
Llama ZS	unit ambiguity	30	0.0	0.0	100.0	
Llama ZS	missing parameter	40		0.0	100.0	
Llama ZS	multi crop	30				
Llama FT 42	seed	182	88.2			100.0
Llama FT 42	code switched	40	50.0			
Llama FT 42	misspelled crop	40	66.7	0.0	100.0	
Llama FT 42	unit ambiguity	30	61.1	0.0	100.0	
Llama FT 42	missing parameter	40		0.0	100.0	
Llama FT 42	multi crop	30				
Llama CL 42	seed	182	85.3			100.0
Llama CL 42	code switched	40	60.0			
Llama CL 42	misspelled crop	40	66.7	0.0	100.0	
Llama CL 42	unit ambiguity	30	55.6	25.0	75.0	
Llama CL 42	missing parameter	40		50.0	45.0	
Llama CL 42	multi crop	30				
Mistral FT 42	seed	182	85.3			100.0
Mistral FT 42	code switched	40	62.5			
Mistral FT 42	misspelled crop	40	45.8	0.0	100.0	
Mistral FT 42	unit ambiguity	30	50.0	0.0	100.0	
Mistral FT 42	missing parameter	40		0.0	100.0	
Mistral FT 42	multi crop	30				

Appendix D

Rating Instructions

Annotators received a blinded workbook, `reports/s7/stage7_rating_sheet.xlsx`, with one row per judgement. The model name was not in the sheet. The README sheet of that workbook is the instruction they were given. It is reprinted below, then the 1–5 rule that the audit report records as the team's working rubric. The two sources do not say the same thing about what 1 and 2 mean; Section 4.4.6 and the threats section treat that drift as a limit on the 1-versus-2 distinction.

Text of the rating-sheet README

> STAGE 7 HUMAN RATING SHEET — 800 judgements (200 items × 4 model outputs).
>
> Each row = one question + one model's answer. The model is not identified. The same question appears four times in the sheet, in shuffled positions.
>
> Fill only the yellow columns:
>
> - `rating_1_5` — integer 1 (unusable) .. 5 (correct, complete, safe to give)
> - `would_give_farmer` — yes / no — would you hand this answer to a farmer as-is?
> - `notes` — optional, one line
>
> Do not edit any other column. Do not sort or delete rows; `judgment_id` is the join key. `gold_answer` is the reference answer for the question (hide the column if you want to rate blind to the reference).

Working 1–5 rule used in the audit

The Stage 7 audit report records the following working meanings, which are stricter than "unusable" / "safe to give":

- 1 — wrong in a way a farmer would not notice. Invented values are scored 1.
- 2 — wrong in a way a farmer would notice (rambling, unfinished, obviously broken).
- 3 — mixed or partly useful.

- 4–5 — good enough to hand over; asking for a missing input deserves 5.

The gold answer was shown. Annotators were the authors (**AUTHOR INPUT:** confirm annotator identities); none is an agricultural expert. Two cells were corrected after return, as recorded in `reports/human_eval/stage7_audit_report.md`: a blank farmer decision set to yes, and a rating-1 / farmer-yes contradiction set to no.

Appendix E

Consent Form

Written consent was not collected before the authors rated the outputs. A form now exists. For the authors it records, with the real signing date, that the rating was completed before the form was signed. External raters in the supplementary round sign it before they rate (**STAGE 8B, NOT YET RUN:** external raters). The signed forms are **PENDING CONSENT:** signed consent forms.

The text of the form is not in the repository.

AUTHOR INPUT: Paste the consent-form text here, or a pointer to the signed PDF bundle. Until then this appendix is a placeholder.

Appendix F

Detector Rules

The post-hoc detector is implemented in `scripts/s6c_postfix_clarify_detector.py` and applied only to the clarify-trained Llama predictions. It is an oracle-gated diagnostic. It reads gold labels, runs only on known-incomplete queries, and its rules were written with the test categories in view. The Stage 8 result—safe handling 36.8% → 83.8%, risk difference +47.1 [+35.3, +58.8] percentage points, $b = 32$, $c = 0$—is an upper bound on what these surface rules catch in this test set, not a performance estimate.

Scorable clarify rows: 68 (40 missing-parameter, 16 misspelled-crop, 12 unit-ambiguity). The 20 conflicting-value clarify rows are `pending_human` and unscored.

With rules 1 and 3b, every clarify row on which clarify-trained Llama still produced a numeric answer is flagged. The only unflagged unsafe rows are those the classifier did not call an answer, or that matched no unit pattern.

Table F.1: Detector rules as implemented, what each rule reads, and the test categories that informed it.

#	Rule	Reads	Informed by
1	Run only when gold `qtype` is `clarify`; every other row has `detector_flag` false.	gold qtype	All clarify categories. Makes $c = 0$ by construction.
2	`invented_answer` if the response is classed as `answer` and matches a numeric-with-unit pattern (mm / liter / litre / l, or মিমি / লিটার).	generation	The worked-answer template always emits a depth and a volume.
3a	A slot is missing if none of its keywords appears in the question: `et0_mm`, `rainfall_mm`, `area`, `stage` (English and Bengali lists in the script).	question	Training slots `missing_rainfall`, `missing_area`, `missing_stage`. `missing_et0` was added after the test rows were inspected; training never deleted ET_0.
3b	Flag if `invented_answer` and (any slot missing by 3a *or* gold category is missing-parameter, misspelled-crop, or unit-ambiguity).	gold category	Exactly the three scorable clarify categories. The or-branch flags every invented answer on those rows.
4	Correction text: "Please provide: ⟨slot⟩", with the slot taken from the gold subtype.	gold subtype	`ambiguous` → which crop; `unknown_local_unit` → field size and unit; `missing_*` → the missing parameter.

Appendix G

Reproducibility Record

Every number in the body is a macro written by `scripts/thesis/make_numbers.py` from the files listed in `thesis/generated/facts.md`. This appendix records the hashes that lock those files.

The test set used for every Stage 6 and Stage 8 number is `data/hardened/test_hardened.jsonl`. Its SHA-256 is checked against `data/hardened/test_hardened.sha256` (prefix 9314d18153ef) before evaluation and before any thesis script that reads the test set. The seed test split, hashed before hardening, has prefix ad90633b6c26.

Table G.1 lists those two files and the SHA-256 prefix of every adapter loaded at evaluation. Re-verification after the fact found that all 16 of the 16 fine-tuned and clarify-trained arms loaded an adapter whose hash matches its training manifest, even though the evaluation-time manifest check was skipped in 18 runs (Section 4.4.1).

The software freeze used for Stage 8 is `reports/stats/env_freeze.txt`. The analysis plan is `reports/stats/test_plan.json`, version 3. Python, NumPy, SciPy, statsmodels, scikit-learn, pandas and the `krippendorff` package versions are recorded in that plan's `environment` field. R is not available on the build machine.

Training used Unsloth, TRL and Transformers as cited in Chapter 4. Decoding was greedy, with at most 512 new tokens, and 4-bit base weights for every arm.

Table G.1: SHA-256 prefixes (first 12 hex characters) of the locked test files and of every adapter loaded at evaluation. Full digests are in the `.sha256` sidecars and in `thesis/generated/adapter_verification.json`. All 16 adapter arms matched their training manifests.

File / arm	What	SHA-256 prefix	Check
`data/hardened/test_hardened.jsonl`	locked test set	9314d18153ef…	recorded in `test_hardened.sha256`
`data/splits/test.jsonl`	seed test split	ad90633b6c26…	recorded in `test.sha256`
`llama31_8b_seed42`	adapter	cb6d4fe407a3…	matches manifest
`llama31_8b_seed1337`	adapter	70863d04ed30…	matches manifest
`llama31_8b_seed2024`	adapter	653953607991…	matches manifest
`llama31_8b_seed42_clarify`	adapter	bb12b4b54efd…	matches manifest
`mistral_nemo_seed42`	adapter	8a3f5bb0d6e9…	matches manifest
`mistral_nemo_seed1337`	adapter	3400d61c16e8…	matches manifest
`mistral_nemo_seed2024`	adapter	04837f92d2ae…	matches manifest
`mistral_nemo_seed42_clarify`	adapter	5d2b152f15b1…	matches manifest
`phi35_mini_seed42`	adapter	508ea2df2c4b…	matches manifest
`phi35_mini_seed1337`	adapter	0f123ba75443…	matches manifest
`phi35_mini_seed2024`	adapter	c0473932e168…	matches manifest
`phi35_mini_seed42_clarify`	adapter	b0277c35d8bd…	matches manifest
`phi4_mini_seed42`	adapter	925494e55287…	matches manifest
`phi4_mini_seed1337`	adapter	2e603103c348…	matches manifest
`phi4_mini_seed2024`	adapter	c8eb28a72cf3…	matches manifest
`phi4_mini_seed42_clarify`	adapter	40b53437e84c…	matches manifest

Appendix H

Declaration of AI Assistance

The department's required wording is **AUTHOR INPUT:** department's AI-declaration wording. Until that wording is pasted, this appendix records what was actually done.

AI assistants (Claude, Cursor, and related coding agents) were used in this project for three things.

1. **Code.** Writing and revising training, evaluation, leakage, human-study and statistical scripts; generating figures and tables; checking that a number in the text matches a file.
2. **Analysis.** Drafting the Stage 8 test plan and the report generator; the tests themselves were run by `scripts/s8_stats.py` on saved generations. No p-value in the thesis was typed by hand.
3. **Prose.** Drafting and revising chapters, the abstract check, and this appendix. Every factual number in the drafted text is a LaTeX macro expanded from `thesis/generated/numbers.tex`. Every bibliographic entry was checked against its source by `scripts/thesis/verify_refs.py`.

What the assistants did not do: they did not invent gold answers, they did not rate outputs, they did not train a model, and they did not change a locked test row. The authors remain responsible for the claims, the design decisions, and any error that remains.